

Searching for the digital dividend: Internet diffusion and productivity growth in 149 economies, 1996–2025

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Abstract

We examine conditional associations between lagged internet use and labour-productivity growth in 149 economies over 1996–2025. Within countries, years with a higher internet-user share were not years of faster growth. A linear fixed-effects estimate of **−0.284** percentage points per ten-percentage-point higher share is robust to country-specific trends and convergence controls. A cross-fitted machine-learning estimate of **−0.264** is less precise, and its 95% interval includes zero under two-way clustering. The association is concentrated in 1996–2011. To study heterogeneity we use Panel-DOSE, a debiased machine-learning spline estimator for fixed-effects panels. The association does not vary robustly with schooling, initial productivity or time since internet take-off, although the data cannot rule out moderate gradients. A gradient along initial productivity appears in both Penn World Table releases but is much weaker in World Bank data for the same country-years; about half of this difference comes from the growth series, and it is imprecisely estimated. These observational results do not identify the causal return to connectivity investment.

Keywords: Internet diffusion, Labor productivity, Double machine learning, Heterogeneous effects, Panel data, Data revisions

JEL Classification: C14 , C23 , C55 , O33 , O47

1 Introduction

The expectation that digital technologies deliver a “digital dividend” in the form of faster productivity growth is central to development strategies and to much of the

digital-economy literature (World Bank 2016; Goldfarb and Tucker 2019). The empirical record is more mixed than this expectation suggests. Early macro studies of telecommunications and broadband report large growth effects (Röller and Waverman 2001; Koutroumpis 2009; Czernich et al. 2011), and growth accounting attributes much of the late-1990s US productivity revival to information technology (Jorgenson et al. 2008; Jorgenson and Vu 2016). Firm-level studies find large returns mainly where technology is combined with organisational change and skilled labour (Brynjolfsson and Hitt 2000; Bresnahan et al. 2002; Bloom et al. 2012), while cross-country evidence for developing and emerging economies is weaker (Niebel 2018; Vu et al. 2020; Cardona et al. 2013). Recent contributions to the *Journal of Digital Economy* likewise emphasise that the consequences of digitalisation depend on the structure of the economy and on how digital inputs enter production (Banga et al. 2026; Chen and Li 2026).

A common thread is heterogeneity. The returns to a new technology are expected to depend on the capacity to absorb it, proxied by human capital (Nelson and Phelps 1966; Benhabib and Spiegel 1994; Cohen and Levinthal 1990); on distance to the technological frontier (Acemoglu et al. 2006); and on how long the technology has been in use, because general-purpose technologies require complementary intangible investment that depresses measured productivity before raising it (Brynjolfsson et al. 2021). Each argument predicts a specific shape of the function linking the effect of adoption to a characteristic of the adopting economy. Testing these predictions requires an estimator that recovers that function without imposing a linear interaction and without a linear control function, because the determinants of productivity growth and of adoption—income, demography, openness, urbanisation—interact nonlinearly.

This paper examines whether, over three decades of internet diffusion, a productivity dividend is visible in aggregate data, and whether the conditional association between adoption and growth varies in the ways these theories predict. We do not claim to identify the causal effect of adoption: the design is observational, and the estimates are associations conditional on controls and fixed effects. We combine World Bank productivity data and International Telecommunication Union (ITU) adoption data for 149 economies over 1996–2025 with schooling data from the United Nations Development Programme (UNDP), and we compare the results with two releases of the Penn World Table (PWT) (Feenstra et al. 2015; Groningen Growth and Development Centre 2025). To estimate how the association varies with a moderator we use Panel-DOSE (Panel Debiased Orthogonal Sieve Estimator), which approximates the varying coefficient by a penalised spline and estimates it from Neyman-orthogonal moments with machine-learning nuisance functions, Mundlak features, cross-fitting by country and a two-way within transformation.

Three findings stand out. First, within countries, a higher lagged internet-user share is not associated with faster labour-productivity growth. A linear fixed-effects regression gives -0.284 percentage points (pp) per ten-percentage-point higher share, and this estimate is stable when we add country-specific trends, lagged productivity or development-specific year effects. The flexible estimate is -0.264 pp, but its interval includes zero with two-way clustering and for about a third of the random seeds of the sample splits, so we do not claim a significant negative association. The association is concentrated in 1996–2011, is weaker without the transition economies, and is

shared with the co-diffusing mobile and broadband series. By contrast, a cross-country long difference shows that economies with larger increases in internet use grew faster between 2000 and 2024, a comparison that fixed effects remove and that is consistent with faster-growing economies adopting faster. Second, we find no robust variation of the within-country association with schooling, with initial productivity or with the time elapsed since internet take-off. None of the three primary tests rejects, but the minimum detectable gradients are moderate, so the data cannot rule out complementarities of that size. Third, estimates of heterogeneity depend on the productivity data. With the same estimator, the gradient along initial productivity is large in both releases of the Penn World Table and much weaker in World Bank data. On the country-years common to both sources, about half of the difference comes from the growth series and a third from the control set, but a paired bootstrap cannot exclude that the sources agree. The revision from PWT 10.0 to PWT 11.0 does not change the gradient detectably. This complements the evidence of [Johnson et al. \(2013\)](#) that the choice of data changes growth results.

The paper also makes a methodological contribution, which we state modestly. Panel-DOSE combines known components: orthogonal series estimation of causal functions ([Semenova and Chernozhukov 2021](#)), the correlated-random-effects and within-group variants of panel double/debiased machine learning (DML) ([Clarke and Polsell 2026](#)), cross-fitting by clusters, and O’Sullivan penalised splines ([O’Sullivan 1986](#); [Wand and Ormerod 2008](#)). We show that the combination is Neyman-orthogonal, that the within step makes the moment invariant to additive country or year errors in the outcome nuisance, that when the moderator is a control only two nuisance functions are needed whatever the flexibility of the sieve, and that the tests of constancy and linearity remain valid for a fixed penalty. Relative to the panel heterogeneous-effect estimator of [Semenova et al. \(2023\)](#), which targets a finite-dimensional projection of the conditional effect under weak time dependence, Panel-DOSE targets a nonparametric function of a single moderator with a continuous treatment under two-way fixed effects and cluster sampling; relative to [Klosin and Vilgalys \(2022\)](#) and [Colanigelo and Lee \(2026\)](#), who estimate average partial effects or dose-response functions, it targets effect heterogeneity. Monte Carlo experiments, including a plasmode design built on the empirical panel for each moderator ([Franklin et al. 2014](#)), show where the estimator helps and where it does not. In the stylised designs it has the lowest integrated error among the feasible estimators we compare when the effect varies and the confounding is strongly nonlinear, and most of the gain comes from the Mundlak features. A two-way fixed-effects (TWFE) regression with spline controls does as well when the confounding is only mildly nonlinear. Linear interaction models produce spurious gradients in the plasmode design built on our data, and the penalised bands under-cover curved effect functions, so we use them descriptively.

Section 2 reviews the literature and states the predictions we test. Section 3 presents the estimator. Section 4 reports the simulations and Section 5 the application. Section 6 discusses interpretation, policy and limitations, and Section 7 concludes. The appendices contain proofs (Appendix A), data and implementation details (Appendix B) and additional results (Appendix C).

2 Related literature and predictions

2.1 ICT, broadband and aggregate growth

Röller and Waverman (2001) estimate a structural model of telecommunications investment and output for OECD countries and find large growth effects of network roll-out; Koutroumpis (2009) obtains similar results for broadband. Czernich et al. (2011) instrument broadband penetration with pre-existing voice and cable networks and find that a ten-point increase raised annual per-capita growth by 0.9–1.5 pp in OECD countries. Growth accounting documents large contributions of ICT capital in advanced economies and, increasingly, in emerging economies (Jorgenson et al. 2008; Jorgenson and Vu 2016). Surveys of the ICT–productivity literature conclude that estimated returns are positive on average but heterogeneous and sensitive to the level of aggregation and to method (Cardona et al. 2013; Vu et al. 2020). Niebel (2018) finds no larger returns to ICT capital in developing than in developed economies. For the developing world, Hjort and Poulsen (2019) show that the arrival of submarine cables raised employment in Africa, and Hjort and Tian (2025) review a literature that finds positive effects of connectivity on employment and firm outcomes but little direct evidence on aggregate productivity.

2.2 Complementarities, diffusion and capabilities

At the firm and worker level, the productivity gains from digital technologies depend on skills and organisation. Broadband raised the productivity of skilled workers and lowered that of unskilled workers in Norway (Akerman et al. 2015); US multinationals obtained higher returns to IT than comparable European firms because of their management practices (Bloom et al. 2012). OECD work finds that digital adoption and its productivity pay-off are concentrated in firms and industries with the capabilities to use it (Gal et al. 2019; Nicoletti et al. 2020). At the country level, Comin and Hobijn (2010) and Comin and Mestieri (2018) show that adoption lags have fallen but the intensity of use still differs widely across countries, so that access and use diverge. Recent contributions to the *Journal of Digital Economy* study how digitalisation relates to structural change across countries (Banga et al. 2026) and how data and digital inputs can be represented in aggregate production functions (Chen and Li 2026); the productivity channel we study complements the structural-change channel emphasised by Banga et al. (2026). Machine-learning methods are also increasingly used in the *Journal of Digital Economy* to predict firm outcomes (Xu et al. 2026); we use them instead to control flexibly for confounders when estimating a heterogeneous association.

2.3 Measurement and data revisions

Cross-country growth regressions are sensitive to the release of the data. Johnson et al. (2013) show that growth rates in successive PWT releases differ substantially for the same countries and years, and that the revisions change the results of growth regressions; Ponomareva and Katayama (2010) show the same for the growth–volatility

relationship. Mismeasurement of digital output has also been proposed as an explanation for the post-2005 productivity slowdown, although it accounts for only a small part of it (Byrne et al. 2016; Syverson 2017). Our vintage comparison applies this insight to heterogeneous associations of internet diffusion and attributes the change between releases to the revised outcome, moderator and controls.

2.4 Machine learning for heterogeneous and continuous effects

Double/debiased machine learning (Chernozhukov et al. 2018) generalises the partialling-out estimator of Robinson (1988). Clarke and Polselli (2026) extend it to static panels with fixed effects through correlated-random-effects, within-group and first-difference implementations of a constant coefficient. Semenova et al. (2023) estimate heterogeneous effects in high-dimensional dynamic panels under weak dependence, targeting the best linear predictor of the conditional effect in a set of covariates. Semenova and Chernozhukov (2021) project orthogonal signals on series bases to estimate causal functions. For continuous treatments, Colangelo and Lee (2026) estimate dose-response functions and partial effects with kernel-based DML, and Klosin and Vilgalys (2022) estimate average partial derivatives in panels with unit effects. Causal forests and the R-learner estimate conditional effects in cross-sectional data (Wager and Athey 2018; Athey et al. 2019; Nie and Wager 2021). The TWFE literature shows that linear fixed-effects regressions with heterogeneous or continuous treatments estimate weighted averages whose weights can be negative or hard to interpret (de Chaisemartin and D’Haultfoeuille 2020; Callaway et al. 2024). Panel-DOSE targets a different estimand from these methods: the function $\theta(z)$ describing how the association of a continuous treatment with the outcome varies with a scalar moderator, under two-way fixed effects and dependence within countries. Its components are known; what the paper adds is their combination for this estimand, the associated properties, and evidence on when the combination helps.

2.5 Predictions

We translate the theories into predictions on the shape of $\theta(\cdot)$ in model (1) below. Absorptive capacity (Nelson and Phelps 1966; Benhabib and Spiegel 1994; Akerman et al. 2015) predicts that θ increases with schooling. Leapfrogging arguments, in which the internet lowers information and transaction costs most where they are highest (Hjort and Poulsen 2019), predict that θ decreases with initial productivity; frontier-based arguments (Acemoglu et al. 2006) allow either sign. The productivity J-curve (Brynjolfsson et al. 2021) predicts that θ is negative in the first years after a country’s adoption takes off and rises with time since take-off. Network effects predict a marginal effect of adoption $f'(d)$ that rises with the level of adoption; saturation predicts that it falls. Finally, if the gains accumulate slowly, cumulative associations over several years should exceed the one-year association. Because our design is observational, we test these predictions on the shape of the conditional association, under the assumptions of Section 3; a failure to find a predicted shape is informative only up to the precision we report.

3 Methodology

In words, the estimator proceeds in four steps. It first predicts productivity growth and internet adoption from each country’s characteristics and their country averages, using a machine-learning model trained on other countries. It then keeps the prediction errors—the parts of growth and adoption that the characteristics do not explain—and removes country and year averages from them. Finally, it fits a smooth curve that describes how the unexplained part of growth co-moves with the unexplained part of adoption for different values of a moderator such as schooling. The height of this curve at a given value of the moderator, $\theta(z)$, is the difference in annual growth, in percentage points, associated with a unit (100 percentage points) higher lagged internet-user share; one tenth of it is the difference associated with ten more users per hundred people. Readers interested mainly in the results can skip to Section 5.

3.1 Model and target parameters

We observe a panel $\{(Y_{it}, D_{it}, X_{it}) : i = 1, \dots, G; t = 1, \dots, T_i\}$ of G countries, where Y_{it} is an outcome (productivity growth), $D_{it} \in \mathbb{R}$ is a continuous treatment (the lagged internet-user share) and $X_{it} \in \mathbb{R}^p$ collects controls that include the moderator Z_{it} . The varying-coefficient model is

$$Y_{it} = \theta(Z_{it}) D_{it} + g(X_{it}) + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (1)$$

in which $g(\cdot)$ is an unknown control function and (α_i, λ_t) are country and year effects. Under the assumptions below, $\theta(z)$ is the difference in Y associated with a unit difference in D for an economy with moderator value z , holding the controls and fixed effects constant. We call it an effect for brevity when discussing the estimator, but in the application it is a conditional association (Section 6). The dose-response variant replaces $\theta(Z_{it})D_{it}$ by $f(D_{it})$ and targets $f'(d)$.

We report four summaries of $\theta(\cdot)$. The *headline average* is the constant-effect coefficient $\bar{\theta}_w$ obtained when the sieve has a single basis function, i.e. the fixed-effects DML (FE-DML) estimator of [Clarke and Polsell \(2026\)](#). It equals a weighted average of $\theta(Z_{it})$ with weights proportional to the residual variance of D_{it} given the controls. The *sample-average effect* is $\bar{\theta} = \mathbb{E}[\theta(Z_{it})]$, which weights country-years equally. The two coincide when θ is constant and differ otherwise; we report both and treat $\bar{\theta}_w$ as the headline because it does not depend on the sieve. The *group-average effects* $\bar{\theta}_A = \mathbb{E}[\theta(Z_{it}) \mid Z_{it} \in A]$ use the terciles of Z , and the *best linear projection (BLP) slope* is $\gamma = \text{Cov}(\theta(Z_{it}), Z_{it}) / \text{Var}(Z_{it})$, computed over the estimation sample.

3.2 Identification

Write $F_{it} = (X_{it}, \bar{X}_i, \bar{D}_i, \bar{Z}_i, t)$ for the controls augmented by their country averages and a time index.

Assumption 1 (Conditional strict exogeneity) $\mathbb{E}[\varepsilon_{it} \mid D_{i1}, \dots, D_{iT_i}, X_{i1}, \dots, X_{iT_i}, \alpha_i] = 0$ for all t .

Assumption 2 (Correlated random effects in the nuisance step) $\mathbb{E}[\alpha_i | F_{it}] = h(\bar{X}_i, \bar{D}_i, \bar{Z}_i)$ for some function h that does not depend on t .

Assumption 3 (Nuisance sufficiency) $\mathbb{E}[D_{it} | F_{i1}, \dots, F_{iT_i}] = \mathbb{E}[D_{it} | F_{it}]$ for all t .

Assumption 1 is the strict-exogeneity condition of static fixed-effects models. It fails mechanically if the controls contain lagged levels or growth of the outcome variable: with $Y_{it} = \Delta \ln \text{LP}_{it}$, the control $\ln \text{LP}_{i,t}$ that enters at $t + 1$ contains Y_{it} , and so does its country mean. Our baseline therefore excludes lagged productivity levels and lagged growth, uses initial (pre-sample) productivity instead, and forms Mundlak averages only of controls that are not functions of the outcome. Assumption 2 is a nonparametric version of the Mundlak device (Mundlak 1978; Wooldridge 2010): the conditional mean of the country effect given the augmented controls may be any function of the country averages. The part of α_i it leaves unexplained, $a_i = \alpha_i - h(\cdot)$, is removed by the within transformation. Assumption 3 requires that the augmented controls of period t summarise the information in the whole control path that is relevant for predicting adoption; the Mundlak averages are included to carry that country-level information. Finally, D must vary after conditioning on F and removing country and year means; we report this residual variation in Section 5.

Let $b(z) = (b_1(z), \dots, b_K(z))'$ be a cubic B-spline basis with interior knots at quantiles of Z , write $\theta(z) = b(z)'\beta + r_K(z)$, define the sieve regressors $\Phi_{it} = D_{it} b(Z_{it})$, and the nuisance functions $\ell(F) = \mathbb{E}[Y | F]$ and $m(F) = \mathbb{E}[\Phi | F]$. Let $\mathcal{M}$ be the two-way within operator (for unbalanced panels, computed by alternating projections). The population moment is

$$\psi(\beta; \ell, m) = \mathbb{E} \left[\sum_t \mathcal{M}(\Phi - m(F))_{it} \left\{ \mathcal{M}(Y - \ell(F))_{it} - \mathcal{M}(\Phi - m(F))'_{it} \beta \right\} \right] = 0. \quad (2)$$

Under Assumptions 1–3 and $r_K = 0$, $Y - \ell(F) - (\Phi - m(F))'\beta = a_i + \varepsilon_{it}$, because λ_t is a function of the time index in F . The term a_i is removed by $\mathcal{M}$ and ε_{it} is mean-independent of the whole path of F , so the true β solves (2).

3.3 Properties

Proposition 1 (Orthogonality and invariance) *Let Assumptions 1–3 hold. (i) Neyman orthogonality: for any directions (δ_ℓ, δ_m) , $\partial_r \psi(\beta_0; \ell_0 + r\delta_\ell, m_0 + r\delta_m)|_{r=0} = 0$. (ii) Invariance for the outcome nuisance: for any country-specific and year-specific constants c_i, c_t , $\psi(\beta; \ell + c_i + c_t, m) = \psi(\beta; \ell, m)$ for all β . The same holds for additive errors $d_i + d_t \in \mathbb{R}^K$ in m . When $m = b(Z)\mu$ (Proposition 2), an error c_i in μ enters the regressors as $\mathcal{M}\{b(Z_{it})c_i\}$; this term is zero if $b(Z_{it})$ is constant within each country (in particular if $K = 1$ or Z is time-invariant), and in general it is not zero otherwise.*

Part (i) makes the estimator insensitive to first-order errors in the machine-learning fits. Part (ii) says that any error in the outcome learner that takes the form of a country or year shift—a natural failure mode when trees approximate fixed effects through

country averages—has no effect on the estimate. For the treatment nuisance the invariance is guaranteed for the constant-effect estimator and for time-invariant moderators; with a time-varying moderator, country-level errors in $\hat{\mu}$ can enter, although only at second order by part (i). A numerical check on the empirical sample confirms both statements (Appendix B).

Proposition 2 (Structure-aware residualisation) *If Z_{it} is an element of F_{it} , then $m_k(F) = b_k(Z) \mu(F)$ with $\mu(F) = \mathbb{E}[D | F]$. Hence the orthogonalised sieve regressors are $b(Z_{it})\{D_{it} - \mu(F_{it})\}$, and only ℓ and μ need to be learned, whatever the sieve dimension K .*

Proposition 2 follows from $\mathbb{E}[D b_k(Z) | F] = b_k(Z) \mathbb{E}[D | F]$. It reduces the number of learners from $K + 1$ to two, avoids fitting K separate regressions for highly collinear spline regressors, and allows the sieve (number of knots, penalty, or a linear basis) to be changed after the nuisance step without refitting. It also implies that the constant-effect FE-DML estimate is obtained from the same residuals. In the dose-response model, Φ is a basis in D and the K conditional means are learned separately.

3.4 The Panel-DOSE algorithm

The final stage minimises a penalised least-squares criterion in the orthogonalised variables,

$$\hat{\beta}_\lambda = \arg \min_{\beta} \sum_{i,t} (\tilde{Y}_{it} - \tilde{\Phi}'_{it} \beta)^2 + \lambda \kappa \beta' \Omega \beta, \quad \Omega = \int b''(z) b''(z)' dz, \quad (3)$$

where $\tilde{Y} = \mathcal{M}(Y - \hat{\ell})$, $\tilde{\Phi} = \mathcal{M}(\Phi - \hat{m})$, Ω is the O’Sullivan penalty (O’Sullivan 1986; Wand and Ormerod 2008) and $\kappa = \text{tr}(\tilde{\Phi}' \tilde{\Phi}) / \text{tr}(\Omega)$ makes λ scale-free. Because $\beta' \Omega \beta$ is the integrated squared second derivative of $b(z)' \beta$, its null space consists exactly of the linear functions, whatever the knot placement, so a large λ shrinks θ towards a linear function.¹ We choose λ by ten-fold cross-validation in which the folds are formed by countries, over $\lambda \in \{0\} \cup \{10^{-6}, \dots, 10^4\}$, and report the selected λ and the effective degrees of freedom $\text{edf} = \text{tr}\{(\tilde{\Phi}' \tilde{\Phi} + \lambda \kappa \Omega)^{-1} \tilde{\Phi}' \tilde{\Phi}\}$. At the upper end of the grid the fit is effectively linear ($\text{edf} \approx 2$). To avoid reporting a median over splits that selected very different amounts of smoothing, we choose one λ for all splits, the minimiser of the cross-validation criterion summed over the S splits. Algorithm 1 in the supplementary material summarises the procedure.

Cross-fitting by countries rather than by observations is essential in a panel. The Mundlak means are computed within each country, so with folds formed by countries no feature of a training country contains information about a test country; using the country mean of D , which includes future adoption, is justified by strict exogeneity (Assumption 1). With observation-level folds, the learner is trained on other years of the same country and partly reproduces its own country effect. Repeating the split S times and aggregating by the median, functional by functional, with the variance

¹The second-difference penalty of Eilers and Marx (1996) has this property only for equally spaced knots; with knots at quantiles of a skewed moderator its null space contains nonlinear functions.

adjustment of Chernozhukov et al. (2018), reduces the sensitivity of the results to a particular partition. Any linear transformation of the functionals—centring or detrending a grid, taking a contrast—is applied to each split’s coefficients before the median is taken, so that every reported number is aggregated in the same way. The element-wise median of the covariance matrices is projected on the positive semi-definite cone; its diagonal equals the median-aggregated variances. Because the median of a difference need not equal the difference of the medians, a reported contrast can differ slightly from the difference of the reported group averages.

3.5 Inference

Let $A_\lambda = \tilde{\Phi}'\tilde{\Phi} + \lambda\kappa\Omega$ and $\hat{u}_{it} = \tilde{Y}_{it} - \tilde{\Phi}'_{it}\hat{\beta}$. The cluster-robust covariance estimator is

$$\hat{V} = c_n A_\lambda^{-1} \left(\sum_{i=1}^G \hat{s}_i \hat{s}_i' \right) A_\lambda^{-1}, \quad \hat{s}_i = \sum_t \tilde{\Phi}_{it} \hat{u}_{it}, \quad (4)$$

with $c_n = \frac{G}{G-1} \frac{n-1}{n-K}$ (Cameron and Miller 2015). Standard errors of linear functionals $w'\beta$ follow directly. Simultaneous bands over a grid use the sup- t critical value from the correlation matrix of the estimated function (Montiel Olea and Plagborg-Møller 2019). We test whether θ is constant, and whether it is linear, with sup- t statistics on a 25-point grid of moderator quantiles. The statistic is the maximum absolute t -ratio of the centred (or linearly detrended) function, and its null distribution is simulated from the corresponding, possibly rank-deficient, covariance matrix, so no matrix inversion is required.² For each moderator, the constancy test is the primary test of heterogeneity, and we adjust the three primary tests (one per moderator) by the method of Holm (1979). We designate these tests as primary in this paper; there is no registered pre-analysis plan. Tercile contrasts and BLP slopes are descriptive summaries, and robustness checks are not treated as additional hypotheses. To compare two estimates computed on the same countries, such as those from the two PWT releases, we use a paired country bootstrap that re-estimates the nuisance functions, the penalty and the tercile groups of both estimates in every draw. We also report two-way (country and year) clustered standard errors for the baseline, $c_G V_{\text{country}} + c_T V_{\text{year}} - c_n V_{\text{obs}}$ with $c_G = G/(G-1)$, $c_T = T/(T-1)$ and $c_n = n/(n-K)$ (Cameron et al. 2011), replaced by the country-clustered variance if the combination is negative. With about 30 years, the year dimension provides few clusters, so these standard errors are themselves imprecise.

Assumption 4 (Regularity) (a) Countries are independent, $G \rightarrow \infty$, and $T_i \leq \bar{T} < \infty$. (b) The cross-fitted nuisance estimators satisfy $\|\hat{\mu} - \mu\|_2 (\|\hat{\mu} - \mu\|_2 + \|Q(\hat{\ell} - \ell)\|_2) = o_p(G^{-1/2})$ and $\|\hat{\ell} - \ell\|_2 = o_p(1)$, where Q removes country means: by Proposition 1(ii), only the within-country error of $\hat{\ell}$ matters, whereas the full error of $\hat{\mu}$ enters. (c) $J = \mathbb{E}[\sum_t \tilde{\Phi}_{it}^0 \tilde{\Phi}_{it}^{0'}]$ is positive definite and fourth moments are bounded. (d) K is fixed and $\lambda_G \sqrt{G} \rightarrow 0$.

²Wald tests on differences of spline coefficients would require inverting a near-singular covariance matrix; the sup- t statistics avoid this.

Proposition 3 (Asymptotic normality) *Under Assumptions 1–4, Assumption 5 in A, and the conditions of Proposition 2, $\sqrt{G}(\hat{\beta} - \beta_K) \xrightarrow{d} N(0, J^{-1}\Omega_s J^{-1})$, where β_K is the sieve pseudo-true value and $\Omega_s = \mathbb{E}[s_i s_i']$ with $s_i = \sum_t \tilde{\Phi}_{it}^0 u_{it}$, and $G\hat{V} \xrightarrow{p} J^{-1}\Omega_s J^{-1}$.*

Proposition 3 requires a vanishing penalty. The cross-validated penalty of the application does not vanish: κ grows with the number of observations, so that a fixed λ corresponds to a penalty of order G in (3). The next result shows that the shape tests remain valid for any fixed penalty, because constant and linear functions lie in the null space of the O’Sullivan penalty.

Proposition 4 (Shape tests under a non-vanishing penalty) *Let the conditions of Proposition 3 hold except that $\lambda \geq 0$ is fixed, and let $\bar{\kappa} = \text{plim } \kappa/G$ and $J_\lambda = J + \lambda \bar{\kappa} \Omega$. Then $\sqrt{G}(\hat{\beta}_\lambda - \tilde{\beta}_\lambda) \xrightarrow{d} N(0, J_\lambda^{-1}\Omega_s J_\lambda^{-1})$, where $\tilde{\beta}_\lambda = (\tilde{\Phi}'\tilde{\Phi} + \lambda \bar{\kappa} \Omega)^{-1} \tilde{\Phi}'\tilde{\Phi} \beta_K$ is the penalised projection of the sieve coefficients. If θ is linear in the winsorised moderator, then $\Omega \beta_K = 0$, so $\tilde{\beta}_\lambda = \beta_K$: the penalised estimator has no smoothing bias, $G\hat{V} \xrightarrow{p} J_\lambda^{-1}\Omega_s J_\lambda^{-1}$, and the sup- t linearity test has asymptotic size equal to its nominal level. The same holds for the constancy test when θ is constant, and, in the dose-response model, for the test that f' is constant (linear f).*

Remark 1 Propositions 3 and 4 are fixed- K results. Proposition 3 gives inference on the pseudo-true function $b(z)'\beta_K$ and on its functionals—group averages, contrasts and the BLP slope—not on $\theta(z)$ itself. With a non-vanishing penalty, Proposition 4 centres the estimator at the penalised projection $\tilde{\beta}_\lambda$, which depends on the sample design matrix; off the null, intervals for penalised functionals are therefore conditional on the design and omit the variability of $\tilde{\beta}_\lambda$ around its limit, and they do not cover a curved θ at the nominal rate. Two further qualifications apply. First, the moderator is winsorised at its 0.5th and 99.5th percentiles before the basis is formed, so the null hypotheses concern θ on that range (a linear θ is linear in the winsorised moderator). Second, the theory treats λ as fixed, whereas the application selects it by cross-validation. We therefore calibrate the tests by simulation (Section 4.3), report their simulated size alongside the p -values, and read the bands as descriptions of the penalised fit rather than as confidence sets for θ . Inference on $\theta(z)$ with $K \rightarrow \infty$ would require the conditions of Belloni et al. (2015) and Semenova and Chernozhukov (2021), which we do not verify for panels.

4 Monte Carlo evidence

4.1 Stylised design

The first design mimics a macro panel of $G = 100$ countries observed for $T = 24$ years. Each country has a fixed effect $\alpha_i \sim N(0, 1)$ and there is a random-walk year effect. Five controls follow $X_{it} = 0.6X_{i,t-1} + 0.4(\eta_{it} + 0.5\alpha_i)$ with $\eta_{it} \sim N(0, I_5)$. The moderator $Z_{it} \in [0, 1]$ trends slowly around a country-specific level that is correlated with the country effect through $\mathbb{I}\{\alpha_i > 0\}$. Adoption follows a logistic diffusion index,

$$D_{it} = \Lambda(-3 + 0.22t + 0.6\alpha_i + 0.8 \sin(1.5X_{1it}) + 0.6X_{2it}\mathbb{I}\{X_{3it} > 0\} + 1.5Z_{it} + v_{it}),$$

where Λ is the logistic function and $v_{it} \sim N(0, 1)$, and the outcome is $Y_{it} = \theta(Z_{it})D_{it} + g(X_{it}, Z_{it}) + \alpha_i + \lambda_t + \varepsilon_{it}$ with $\varepsilon_{it} \sim N(0, 0.25)$ and

$$g = \rho\{1.5 \sin(1.5X_{1it}) + X_{2it}\mathbb{I}\{X_{3it} > 0\} + 0.5|X_{4it}| + 3.2(Z_{it} - 0.5)^2\}.$$

The four effect functions are *constant* $\theta(z) = 1$, *linear* $-0.5 + 3z$, *threshold* $-0.5 + 2.5/(1 + e^{-12(z-0.5)})$ and *hump* $2 - 8(z - 0.5)^2$. The baseline sets $\rho = 1$; for the threshold design we vary the confounding strength ($\rho \in \{0.25, 0.5, 2\}$) and the number of countries ($G \in \{50, 200\}$). This design favours flexible estimators by construction, since the confounding is strongly nonlinear and the residual variation in adoption is larger than in the application; the plasmode design below addresses this.

We compare twelve estimators: (1) TWFE with a constant effect; (2) TWFE with $D \times Z$ and $D \times Z^2$; (3) TWFE-spline, with additive spline expansions of every control and a spline in Z for the effect; (4) FE-DML with a constant effect (Clarke and Polselli 2026); (5) a boosted R-learner (Nie and Wager 2021) fitted to the FE-DML residuals, with its hyper-parameters chosen by country-level cross-validation; (6) a within-group DML sieve, which learns the nuisance functions on two-way demeaned data, in the spirit of the within-group estimator of Clarke and Polselli (2026); (7) a pooled DML sieve without Mundlak features or within step; (8) an ablation with Mundlak features but no within step; (9) an ablation with the within step but no Mundlak features; (10) Panel-DOSE without penalty; (11) Panel-DOSE; and (12) a latent-information oracle that replaces the learned nuisance functions by the conditional means of D and Y given the controls and the true country and year effects. The oracle uses information that no feasible estimator has; it is not the population version of the nuisance problem that Panel-DOSE solves, whose features contain Mundlak means rather than α_i , and we use it only as a benchmark. All machine-learning estimators use LightGBM (Ke et al. 2017) with 600 trees, five country-level folds and two repetitions of the split with a pooled penalty; the sieve estimators use four interior knots. We replicate each design 100 times (50 in the sensitivity analysis) and report the integrated root-mean-squared error (IRMSE) over 17 grid points in $[0.1, 0.9]$, the integrated bias, the RMSE of the high-minus-low tercile contrast, and the coverage of nominal 95% intervals with Monte Carlo standard errors.

4.2 Results of the stylised design

Table 1 (and Figure C2 in the supplementary material) reports accuracy, and Table C7 in the supplementary material reports coverage. Linear fixed-effects models are heavily biased because the control function is nonlinear and correlated with adoption. The quadratic interaction does not help: its errors are identical in the constant, linear and hump designs because these effect functions lie in the span of its regressors, so the problem is the control function, not the flexibility of the effect. TWFE with spline controls removes most of the bias but keeps an integrated bias of about 0.23.

Among the feasible flexible estimators, Panel-DOSE has the lowest IRMSE in the three heterogeneous designs (0.15, 0.23 and 0.22 for the linear, threshold and hump designs), 20–47% below TWFE-spline and 15–45% below the tuned R-learner. The constant-effect FE-DML estimator is best when the effect is in fact constant. The

ablations locate the gain in these designs. Dropping the Mundlak features multiplies the error by 2.1 to 2.9 (0.45–0.52), whereas dropping the within step raises it by 7–16% (0.17–0.25). Learning the nuisance on demeaned data (the within-group sieve) performs poorly (0.70–0.92), because $g(X)$ minus its country mean is not a function of the demeaned controls when g is nonlinear. This is a structural limitation of that benchmark under a nonlinear control function, not evidence that its learner is weaker; similarly, the R-learner works with scalar transformed residuals rather than with demeaned interaction regressors. The comparison therefore shows the value of the Mundlak features for this estimand and these designs, not a general superiority over panel DML methods built for other estimands. The latent-information oracle has 16–22% lower IRMSE than Panel-DOSE, which indicates how much could be gained from better nuisance estimates in this design.

Coverage is below nominal in some cells. For Panel-DOSE, pointwise coverage is 0.81–0.91 and simultaneous coverage 0.55–0.85. It is worst in the threshold design, where the penalty smooths the steep part of the curve, but it is also below nominal when θ is constant and there is no smoothing bias (0.84 simultaneous), so the data-dependent penalty and the finite number of clusters contribute as well. The unpenalised sieve has pointwise coverage of 0.93–0.94 and simultaneous coverage of 0.83–0.88. Coverage of the smooth functionals is closer to nominal: 0.88–0.92 for the average effect, 0.90–0.97 for the tercile contrast and 0.88–0.94 for the BLP slope. With the oracle nuisance, average-effect coverage is 0.95–0.97. The sup- t constancy test rejects in 0.09 of replications when the effect is constant (0.07 without penalty), somewhat above the nominal 5%, and the linearity test in 0.06 when it is linear (0.05 without penalty); against the threshold and hump designs the constancy test rejects in 1.00 and 0.95. Intervals from the TWFE-interaction model cover the true tercile contrast in only 0.03–0.17 of replications, so its apparent precision is not informative when the control function is misspecified.

The sensitivity analysis (Table C8 in the supplementary material) qualifies the ranking. With strong confounding ($\rho = 2$) or with 50 or 200 countries, Panel-DOSE remains the most accurate estimator. With mild confounding, TWFE-spline is as accurate or more accurate (IRMSE 0.17 against 0.22 at $\rho = 0.25$ and 0.20 against 0.23 at $\rho = 0.5$), so the gain from machine learning is specific to strongly nonlinear confounding. Simultaneous coverage of the penalised bands falls to 0.40 at $\rho = 2$. The cross-validated penalty selects $\lambda = 0$ in 0% and the upper end of the grid (an effectively linear fit) in 33% of replications.

4.3 Plasmode design

The stylised design has less noise and more residual variation in adoption than the application. The plasmode design (Franklin et al. 2014) keeps the actual controls, treatment, moderators and panel structure of the baseline sample of Section 5 (4,075 observations, 149 countries) and regenerates only the outcome, as $Y_{it}^* = \theta(Z_{it})D_{it} + \hat{g}(F_{it}) + w_i\hat{e}_{it}$. Here $\hat{g}$ is a boosted-tree fit of observed productivity growth on the nuisance features, $\hat{e}_{it}$ are the corresponding residuals, and w_i are Rademacher weights drawn per country. The errors thus keep their size, heteroskedasticity and within-country dependence, while any association with D is randomised away. The design

Table 1 Stylised Monte Carlo design: accuracy ($G = 100$, $T = 24$, $\rho = 1$).

Estimator	IRMSE				Integrated bias				RMSE, high–low contrast			
	Const.	Lin.	Thr.	Hump	Const.	Lin.	Thr.	Hump	Const.	Lin.	Thr.	Hump
TWFE	0.63	0.94	1.15	1.05	0.63	0.57	0.60	0.96	0.00	1.41	2.07	0.17
TWFE–interaction	0.83	0.83	0.87	0.83	0.77	0.77	0.66	0.77	0.52	0.52	0.73	0.52
TWFE–spline	0.29	0.29	0.29	0.29	0.23	0.23	0.23	0.23	0.16	0.16	0.16	0.16
FE–DML (constant)	0.07	0.74	0.99	0.49	0.06	0.06	0.07	0.24	0.00	1.41	2.07	0.17
FE–DML + R–learner (tuned)	0.24	0.28	0.28	0.30	0.06	0.06	0.06	0.10	0.16	0.31	0.37	0.19
WG–DML sieve	0.61	0.74	0.70	0.92	0.53	0.45	0.48	0.81	0.40	0.78	0.68	0.31
Pooled DML sieve	0.63	0.64	0.68	0.71	0.60	0.61	0.63	0.66	0.27	0.29	0.30	0.28
Ablation: Mundlak only	0.18	0.17	0.25	0.25	0.10	0.07	0.07	0.14	0.21	0.23	0.25	0.21
Ablation: within only	0.44	0.45	0.49	0.52	0.41	0.41	0.42	0.46	0.24	0.27	0.28	0.25
Panel–DOSE (unpenalised)	0.26	0.26	0.27	0.26	0.05	0.06	0.07	0.05	0.20	0.22	0.22	0.20
Panel–DOSE	0.15	0.15	0.23	0.22	0.05	0.06	0.06	0.08	0.20	0.20	0.22	0.20
Latent–information oracle	0.12	0.12	0.20	0.18	0.00	0.00	–0.00	0.03	0.18	0.18	0.19	0.18

Notes: averages over 100 replications. The lowest IRMSE among feasible estimators in each design is in bold. The latent–information oracle uses the true country and year effects and is infeasible. IRMSE and integrated bias are computed over 17 grid points in $[0.1, 0.9]$.

inherits the limitations of a single pilot fit: $\hat{g}$ is smoother than the true control function, the design holds the panel fixed, and the country–level sign flips remove any correlation of the errors across countries within a year beyond the year effects. For each of the three moderators we consider no effect (the size of the constancy test), a linear gradient (the size of the linearity test and the power of the constancy test) and a smooth threshold (curvature). The linear and threshold functions are scaled so that their tercile contrast equals the minimum detectable contrast of the application, 2.8 times the baseline standard error (10.07, 9.16 and 31.27). The estimator is run as in the application’s robustness checks, with ten repetitions of the split and one cross–validated penalty pooled over the splits, and the pilot fit $\hat{g}$ is itself cross–fitted by country. We use 200 replications under no effect, so that the Monte Carlo standard error of a rejection rate of 5% is 1.5 percentage points, and 100 under each alternative.

Table 2 reports the results. Under no effect, the penalised constancy test rejects in 0.07, 0.09 and 0.06 of the replications for schooling, initial productivity and years since take–off, and the unpenalised test in 0.04, 0.07 and 0.03; with a Monte Carlo standard error of about 1.5 percentage points, the penalised test is slightly over–sized for initial productivity and close to nominal otherwise. The FE–DML test of a zero average association rejects in 0.03–0.06 of the replications, so the headline interval has close to nominal coverage in this design. Under the linear gradient, the penalised linearity test rejects in 0.04, 0.05 and 0.03 of the replications, and the constancy test detects a gradient of the size of the minimum detectable contrast in 0.93, 0.81 and 1.00; without the penalty the power is much lower along initial productivity (0.34). The contrast intervals cover the true contrast in 0.95, 0.96 and 0.94 of the replications under no effect, and the contrast estimates are biased by 0.88, 0.46 and 0.82, about a quarter of their standard errors; this second–order bias of the nuisance step is of the same order as the contrasts estimated in the application, which is another reason not to over–interpret small contrasts. The linearity test has little power against the threshold function (0.17, 0.04 and 0.38), and the simultaneous bands of the penalised fit cover it in 0.80, 0.87 and 0.79 of the replications, which reflects the smoothing bias described in Remark 1. The linear TWFE benchmarks are much less reliable along initial productivity (Table C9 in the supplementary material): with no effect at all, the

Table 2 Plasmode design built on the empirical panel: Panel-DOSE.

True θ	High-low tercile contrast				Test rejection		Band coverage	
	Bias	RMSE	Coverage	Rejection	Constancy	Linearity	Pointwise	Simultaneous
<i>Moderator: Schooling</i>								
no effect	0.88	3.67	0.95 (0.02)	0.05 (0.02)	0.07 (0.02)	0.05 (0.02)	0.93 (0.01)	0.93 (0.02)
no effect, unpenalised	0.59	3.39	0.97 (0.01)	0.03 (0.01)	0.04 (0.01)	0.03 (0.01)	0.95 (0.01)	0.96 (0.01)
linear gradient	2.99	4.46	0.94 (0.02)	0.92 (0.03)	0.93 (0.03)	0.04 (0.02)	0.92 (0.01)	0.87 (0.03)
linear gradient, unpenalised	2.45	3.96	0.97 (0.02)	0.92 (0.03)	0.76 (0.04)	0.03 (0.02)	0.93 (0.01)	0.92 (0.03)
threshold	3.45	4.68	0.92 (0.03)	0.95 (0.02)	0.96 (0.02)	0.17 (0.04)	0.90 (0.01)	0.80 (0.04)
threshold, unpenalised	2.11	3.77	0.96 (0.02)	0.89 (0.03)	0.83 (0.04)	0.07 (0.03)	0.93 (0.01)	0.92 (0.03)
<i>Moderator: Initial productivity</i>								
no effect	0.46	3.15	0.96 (0.01)	0.04 (0.01)	0.09 (0.02)	0.06 (0.02)	0.94 (0.01)	0.94 (0.02)
no effect, unpenalised	0.68	4.49	0.92 (0.02)	0.09 (0.02)	0.07 (0.02)	0.07 (0.02)	0.95 (0.01)	0.94 (0.02)
linear gradient	0.90	3.35	0.98 (0.01)	0.85 (0.04)	0.81 (0.04)	0.05 (0.02)	0.93 (0.01)	0.96 (0.02)
linear gradient, unpenalised	1.30	4.65	0.93 (0.03)	0.60 (0.05)	0.34 (0.05)	0.05 (0.02)	0.95 (0.01)	0.96 (0.02)
threshold	0.38	3.16	0.95 (0.02)	0.81 (0.04)	0.75 (0.04)	0.04 (0.02)	0.92 (0.01)	0.87 (0.03)
threshold, unpenalised	1.38	4.74	0.92 (0.03)	0.61 (0.05)	0.42 (0.05)	0.06 (0.02)	0.96 (0.01)	0.96 (0.02)
<i>Moderator: Years since take-off</i>								
no effect	0.82	7.19	0.94 (0.02)	0.06 (0.02)	0.06 (0.02)	0.04 (0.01)	0.96 (0.01)	0.94 (0.02)
no effect, unpenalised	3.65	13.75	0.98 (0.01)	0.02 (0.01)	0.03 (0.01)	0.04 (0.01)	0.96 (0.01)	0.96 (0.01)
linear gradient	-1.60	6.91	0.98 (0.01)	0.97 (0.02)	1.00 (0.00)	0.03 (0.02)	0.96 (0.01)	0.97 (0.02)
linear gradient, unpenalised	-0.37	12.73	0.98 (0.01)	0.41 (0.05)	0.95 (0.02)	0.03 (0.02)	0.95 (0.01)	0.96 (0.02)
threshold	0.17	7.26	0.96 (0.02)	0.91 (0.03)	1.00 (0.00)	0.38 (0.05)	0.88 (0.02)	0.79 (0.04)
threshold, unpenalised	-0.17	12.69	0.96 (0.02)	0.44 (0.05)	0.97 (0.02)	0.18 (0.04)	0.96 (0.01)	0.95 (0.02)

Notes: Monte Carlo standard errors in parentheses. Under no effect, “Rejection” is the size of the 5% test that the contrast is zero and “Constancy” and “Linearity” are the sizes of the sup- t tests; under a linear gradient, “Linearity” is a size and the other rejection rates are powers. Band coverage refers to the true θ on a 17-point grid between the 5th and 95th percentiles of the moderator.

linear-interaction and spline-control contrasts are biased by 0.59 and 1.05 and reject in 0.11 and 0.33 of the replications.

5 Empirical application: internet diffusion and productivity growth, 1996–2025

5.1 Data and specification

The main data set combines the World Bank’s World Development Indicators (WDI, July 2026 release, retrieved in September 2026) with the UNDP Human Development Report 2025. The outcome is annual growth of GDP per person employed at constant 2021 PPP prices, $Y_{it} = 100 \Delta \ln(\text{GDP}_{it}/\text{EMP}_{it})$, which WDI compiles from national accounts and the modelled employment estimates of the International Labour Organization (ILO) and which is available up to 2025. The treatment is the share of the population using the internet (ITU) in year $t - 1$, measured on $[0, 1]$. It is a lagged *level*, not a change: θ compares growth in years in which a country’s internet share was higher with growth in years in which it was lower, conditional on the controls and on country and year effects, and one tenth of θ is the difference in growth (in pp) associated with a ten-percentage-point higher lagged share. In the baseline, the controls are lagged mean years of schooling, the lagged government consumption share, trade openness, population growth, the age-dependency ratio and the urban population share, and initial log GDP per worker, measured as the average over 1991–1995.

The baseline deliberately excludes three groups of variables used in the earlier literature. Lagged productivity levels and growth are excluded because they violate strict exogeneity mechanically (Section 3). Mobile and fixed-broadband subscriptions

are excluded because they measure the same diffusion process, so that controlling for them turns the estimand into the association of internet use net of co-diffusing technologies. The investment share is excluded because it is a likely channel. We report specifications that add each group back. Variables that enter only such robustness checks do not restrict the baseline sample: an observation is used if the outcome, the treatment and the baseline controls are observed. The sample covers 1996–2025, drops growth observations below the 1st and above the 99th percentile, and keeps countries with at least 15 usable years. Table B1 in B shows the observations lost at each step. We screen the adoption series for implausible values and jumps; the rules flag 39 values, for example a rise in Fiji’s mobile subscriptions from 361 to 574 per 100 people in one year (Appendix B). Flagged values are set to missing, so flags in the mobile and broadband series affect only the specifications that use these series. The resulting panel contains 4,075 observations on 149 countries. It includes 53 of 86 high-income, 45 of 59 upper-middle-income, 37 of 47 lower-middle-income and 14 of 25 low-income economies. Economies are excluded mainly because productivity, schooling or government-consumption data are missing; many of the excluded low-income economies are fragile or conflict-affected states (e.g. Afghanistan, Somalia, South Sudan, Yemen). A stricter sample that also requires the other ICT series, the investment share, lagged productivity and a second lag of the treatment to be observed—the complete-ICT sample (3,787 observations, 141 countries)—is used as a robustness check. Table C3 in the supplementary material reports descriptive statistics.

We also use two releases of the Penn World Table (Feenstra et al. 2015): version 11.0 (Groningen Growth and Development Centre 2025), which covers 118 countries over 1996–2023 in our sample (3,178 observations), and version 10.0, which ends in 2019 (115 countries, 2,658 observations). The outcome is growth of real GDP at constant national prices per person engaged (rgdpna/emp), the variable PWT recommends for growth comparisons. The initial-productivity moderator is instead output-side real GDP at current PPPs per person engaged (cgdpoc/emp), averaged over 1991–1995, which PWT recommends for comparing productive capacity across countries at a point in time; we report the national-prices level as a sensitivity check. PWT 10.0 expresses levels in 2017 and PWT 11.0 in 2021 US dollars. Because the knots and terciles are placed at quantiles and the B-spline basis is invariant to affine transformations of the moderator, a change of base year does not by itself affect the estimates; what matters is how the releases rank countries. The PWT panels use the PWT human-capital index and the same controls as the WDI baseline plus the labour share; PWT also provides total factor productivity (TFP).

We estimate model (1) with three moderators corresponding to the predictions of Section 2.5: mean years of schooling (absorptive capacity), initial log GDP per worker (distance to the frontier and leapfrogging) and years since the internet-user share first reached 10% (the J-curve). Using initial rather than lagged productivity as the development moderator avoids the mechanical link with the outcome and sorts countries rather than country-years into terciles. Years since take-off is computed from the whole adoption path, so before take-off its (negative) value uses the future year in which the share crosses 10%; it describes the stage of the diffusion life cycle but is not

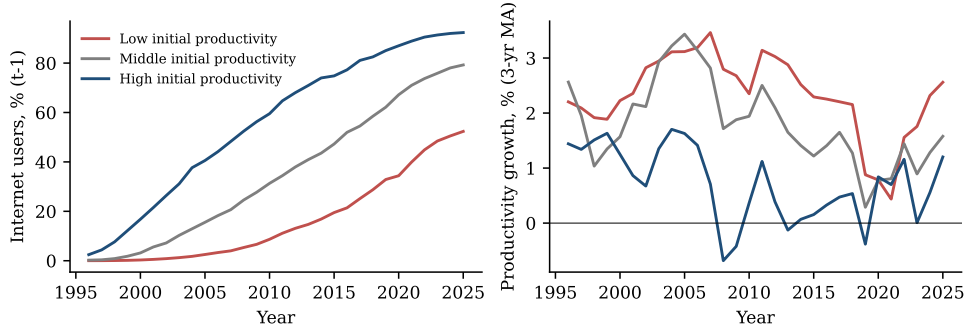


Fig. 1 Internet diffusion and labour-productivity growth by tercile of initial GDP per worker, 1996–2025: (left) sample average of the lagged internet-user share by year; (right) centred three-year moving average of the yearly sample averages of productivity growth.

predetermined. We therefore also use a predetermined version that is zero until take-off and counts the years since take-off afterwards. Panel-DOSE uses LightGBM with 400 trees, five country-level folds, 20 repetitions of the split with one cross-validated penalty pooled over the splits, and cubic B-splines with four interior knots; standard errors are clustered by country. Robustness checks use ten repetitions of the split to save computation, so their FE-DML estimates can differ slightly from the baseline even when the specification is otherwise the same.

Figure 1 shows the variation that identifies the estimates. Internet use diffused everywhere, but at different times. In the countries with the highest initial productivity, about half of the population was online by the mid-2000s, whereas in the least productive third the share passed one fifth only in the mid-2010s and one half at the end of the sample. Smoothed productivity growth was highest in the low and middle groups during the 2000s and fell around 2009 and 2020. Year effects absorb these common shocks, so the estimates rely on differences across countries in the timing of adoption relative to their own productivity paths.

5.2 The average association

In the baseline, the headline FE-DML estimate is -2.64 (s.e. 1.33): a ten-percentage-point higher lagged internet-user share is associated with -0.264 pp lower productivity growth, with a country-clustered 95% interval of $[-0.524, -0.004]$ pp. This interval only just excludes zero, and the conclusion is not robust to reasonable choices. Two-way clustering by country and year gives a standard error of 1.40 and an interval of $[-0.538, 0.009]$ pp. Across 20 random seeds for the sample splits, the estimate ranges from -2.81 to -2.32 (mean -2.61), and the country-clustered interval excludes zero for 65% of the seeds. We therefore read the flexible estimate as negative but not clearly different from zero.

The linear TWFE estimate with the same controls is -2.84 (0.79), and it is stable across checks for trending and convergence-type confounding. It is -3.71 (0.65) with country-specific linear trends, -4.43 (1.24) with quadratic trends, -2.73 (0.74) when

lagged log productivity is added, -2.74 (1.03) with year effects specific to terciles of initial productivity, and -2.26 (0.88) with region-by-year effects (Table C4 in the supplementary material). Mean reversion along a convergence path therefore does not explain the negative sign. The FE-DML estimate is less precise than TWFE because the nuisance step removes most of the variation in adoption. Adoption is highly predictable from the controls, their country means and the year: the out-of-fold R^2 of the treatment nuisance is 0.96 (0.77 within countries), against 0.20 (0.03) for the outcome (Table C5 in the supplementary material). Results are similar across learning rates, tree sizes and numbers of trees, with FE-DML estimates between -2.94 and -2.23 . After the within transformation, only 0.22 of the within-country variance of adoption remains (2.5% of its total variance), and only 0.11–0.14 in the lowest tercile of schooling or initial productivity, where the median adoption share is 0.04–0.07.

The sign of the FE-DML estimate is negative in 32 of the 32 specifications with the baseline outcome and treatment, but only 17 of their country-clustered intervals exclude zero (Figure C1 in the supplementary material). Controlling for mobile and broadband subscriptions reduces the estimate to -0.50 (1.24). These specifications estimate a different quantity—the association of internet use net of co-diffusing technologies—so the fall shows how much of the baseline association the three adoption series share. The estimate is also smaller without the transition economies of Eastern Europe and Central Asia (-1.56 , s.e. 1.29), whose 1991–1995 productivity reflects the post-socialist collapse and whose recovery growth slowed while internet use rose, and without the 8 mainly conflict-affected or fuel-exporting economies that are outside the complete-ICT sample (-2.41 , 1.25). The association is concentrated in the first half of the period. It is -4.91 (2.18) over 1996–2011 and -0.70 (2.15) over 2012–2025; with TWFE it is -4.35 (1.01) and 0.70 (1.46). The 1996–2011 period combines the early diffusion of the internet with the catch-up and commodity boom of the 2000s, whose effects may differ by level of development in ways that the controls do not capture.

The negative association is in output rather than in employment. With TWFE, the lagged internet share is associated with -3.83 (0.78) lower GDP growth and -0.99 (0.33) lower employment growth, so a larger employment denominator does not explain the lower growth of output per worker.

Exogeneity and dynamics

Strict exogeneity implies that future adoption should not predict current growth once current adoption is controlled for (Wooldridge 2010, Section 10.7). Adding the lead $D_{i,t+1}$ to the baseline gives a lead coefficient of -0.87 (1.85) with TWFE and -1.14 (2.06) with FE-DML (Table 3). Neither is significant, but the test has little power because adoption is highly persistent. Used alone, the lead is associated with growth as strongly as the lag (FE-DML -3.63 , s.e. 1.29). In a TWFE profile that dates the share from $t - 6$ to $t + 4$, the coefficients peak around $t - 1$ and t (-3.24 and -2.92) and decline in both directions (-1.81 at $t - 6$, -1.18 at $t + 4$). The association is therefore a low-frequency co-movement between the level of adoption and growth, not a response to changes in adoption: the lagged change $D_t - D_{t-1}$ is associated with growth by -0.37 (1.84) with TWFE and -2.30 (1.76) with FE-DML.

Table 3 and Figure 3(b) report local projections (Jordà 2005) of the cumulative change in log productivity from $t - 1$ to $t + h$ on the internet share in $t - 1$, with the baseline controls; horizon h covers $h + 1$ years of growth. With a persistent regressor, these are close to the cumulated annual association and do not add independent evidence. They show -0.58 pp per ten points over two years ($h = 1$), -1.16 pp over four years and -1.79 pp over nine years, with intervals that include zero from $h = 4$ onwards; on the 2,941 observations for which the nine-year change is observed the pattern is the same. These regressions describe how the conditional association evolves; they do not identify an impulse response to an adoption shock. Within the horizons we observe, they show no delayed positive association. Non-overlapping five-year averages (Islam 1995) give -2.43 (0.84) with TWFE and -2.59 (1.33) with FE-DML.

The picture is different across countries. We regress average annual productivity growth over 2000–2024, i.e. $100(\ln LP_{2024} - \ln LP_{1999})/25$, which is the mean of the annual growth rates used in the panel, on the change in the panel treatment over the same years, $D_{2024} - D_{2000}$ (the internet share of 2023 minus that of 1999), with the controls of 2000 (mostly 1999 values). Economies whose internet share rose more grew faster, by about 0.22 pp per year per ten points (OLS 2.21, s.e. 0.62; DML 2.17, s.e. 1.14; 117 countries). Built from productivity levels without the panel’s trimming and minimum-length rules, the long difference covers 122 countries and gives 2.58 (0.62). Because this comparison does not remove country effects, it is consistent with faster-growing economies adopting faster rather than with adoption raising growth. An instrumental-variable check that instruments adoption with the 1990 density of fixed telephone lines interacted with the global diffusion curve has a weak first stage ($F = 5.6$) and gives an imprecise two-stage least squares (2SLS) estimate of -3.78 (4.00); we report it for completeness only.

Comparison with broadband studies

Our main estimate concerns internet users, per-worker growth and a global panel, whereas Czernich et al. (2011) study broadband, per-capita growth and OECD countries over 1996–2007 with an instrumental-variable design. As a descriptive benchmark closer to their setting, we use fixed broadband subscriptions as the treatment and GDP-per-capita growth as the outcome in high-income economies. The FE-DML estimate is -2.39 (2.55), i.e. $[-0.74, 0.26]$ pp per ten points. Because the populations, periods, treatments and estimands differ, and ours rests on selection on observables, this comparison cannot establish whether the two sets of results are compatible; it shows only that a conditional association of the size they report is not visible in these data. In all countries, the same specification gives -6.35 (2.52), a negative association comparable to that of internet use.

5.3 Heterogeneity

Table 4 and Figure 2 report the heterogeneity estimates. Table C10 in C reports the selected penalties, effective degrees of freedom, identifying variation and treatment support by tercile. The bands in Figure 2 describe the penalised fit (Remark 1); the tests of heterogeneity are the constancy tests.

Table 3 Dynamic specifications and alternative designs.

<i>Panel A. Local projections: change in log productivity ($\times 100$) from $t - 1$ to $t + h$</i>							
Horizon (years)	All available observations			Common sample		High – low	
	Average	(s.e.)	Obs.	Average	(s.e.)	Contrast	(s.e.)
$h = 0$ (1)	–2.71**	(1.29)	4,075	–4.31***	(1.59)	–0.67	(3.23)
$h = 1$ (2)	–5.79**	(2.34)	3,933	–8.85***	(3.04)	0.14	(5.68)
$h = 2$ (3)	–7.84**	(3.70)	3,792	–10.63**	(4.63)	4.01	(9.01)
$h = 3$ (4)	–11.58**	(5.43)	3,649	–12.38**	(6.04)	7.04	(13.88)
$h = 4$ (5)	–11.82*	(6.93)	3,508	–12.33*	(7.49)	13.98	(18.46)
$h = 5$ (6)	–12.66	(8.25)	3,370	–12.86	(8.75)	24.66	(21.98)
$h = 6$ (7)	–11.06	(9.13)	3,230	–14.07	(9.44)	35.31	(28.66)
$h = 7$ (8)	–14.58	(10.43)	3,087	–15.03	(10.26)	42.04	(33.52)
$h = 8$ (9)	–17.87*	(10.71)	2,941	–17.87*	(10.71)	47.33	(37.57)
<i>Panel B. Alternative designs</i>							
Specification	Estimate	(s.e.)	Obs.				
Five-year averages, TWFE	–2.43***	(0.84)	834				
Five-year averages, FE-DML	–2.59*	(1.33)	834				
Long difference, OLS	2.21***	(0.62)	117				
Long difference, DML	2.17*	(1.14)	117				
Lead D_{t+1} , TWFE	–0.87	(1.85)	3,923				
Lead D_{t+1} , FE-DML	–1.14	(2.06)	3,923				
Fixed-line IV (2SLS), first-stage $F = 5.6$	–3.78	(4.00)	3,977				
Broadband, GDP p.c. growth, high income	–2.39	(2.55)	1,545				
Broadband, GDP p.c. growth, all countries	–6.35**	(2.52)	3,995				

Notes: coefficients per unit of adoption (0–1); divide by 10 for a ten-point difference. Panel A: FE-DML average and Panel-DOSE tercile contrast by initial log GDP per worker, baseline controls, ten repetitions of the split (so the $h = 0$ average differs slightly from the 20-split baseline); the horizon in parentheses is the number of annual growth rates covered, $h + 1$. “Common sample”: observations for which the nine-year change is observed. Five-year averages: non-overlapping periods 1996–2000 to 2021–2025, treatment and controls at the start of the period. Long difference: average annual growth of GDP per worker over 2000–2024 on $D_{2024} - D_{2000}$ (internet share 2023 minus 1999), with 2000 values of the controls; OLS with heteroskedasticity-robust (HC1) standard errors and cross-fitted partially linear DML. Lead: coefficient on D_{t+1} when it is added to the baseline (strict-exogeneity check). IV: 2SLS with country and year effects; instrument = 1990 fixed-line density $\times$ leave-one-out global internet share. Broadband rows: FE-DML with fixed broadband subscriptions as treatment and GDP-per-capita growth as outcome. *, **, ***: 10%, 5%, 1%.

Schooling

The association does not vary with schooling. The estimated function is nearly flat (Figure 2a), the high-minus-low tercile contrast is -0.70 (s.e. 3.60), the BLP slope is -0.10 (0.49) per year of schooling, and the constancy test does not reject ($p = 0.96$). The absorptive-capacity prediction of a positive gradient is therefore not supported, but the data cannot rule out moderate gradients either: the 95% interval for the contrast is $[-0.78, 0.64]$ pp per ten points, and the minimum detectable contrast (80% power at 5%) is about 1.01 pp per ten points. A null result for heterogeneity is not evidence against complementarities of this size. The sign of the contrast varies across robustness checks; in the complete-ICT sample it is -3.55 (3.36), and over 1996–2011 it is -12.61 (5.75, constancy $p = 0.06$), against 9.29 (5.08) over 2012–2025. Learning-adjusted years of schooling, which account for the quality of schooling (Angrist et al. 2021), give a similar picture (contrast 0.44, s.e. 3.46).

Initial productivity

Along initial productivity, the WDI estimates show no monotone gradient. The contrast is -0.76 (3.27) and the constancy test gives $p = 0.16$. The estimated function dips in the middle of the distribution (Figure 2b), and in some checks the constancy test rejects at 5%: without the penalty ($p = 0.04$), in the period 1996–2011 ($p = 0.02$, contrast 13.10, s.e. 8.05), without Mundlak features and in a few others in C. These fits

select a small penalty and use four to eight effective degrees of freedom; the penalised test is slightly over-sized for this moderator in the plasmode (0.09 at the 5% level, Section 4.3), and the tercile contrasts of these checks have different signs. We read them as weak evidence of non-monotone variation, not as support for the monotone gradient that the leapfrogging or frontier arguments predict. In the local projections the contrast grows with the horizon, to 47.33 (37.57) at $h = 8$, but it is imprecise at every horizon (Table 3). Excluding the bottom 5% of the moderator gives a contrast of 0.40 (3.15), and using only countries with pre-sample productivity data gives 0.00 (3.20). Distance to the US frontier as the moderator, with terciles formed within year, gives -5.16 (2.88); because this moderator is built from lagged productivity, it is not predetermined, and we report it as descriptive only. The linear benchmarks find a positive gradient: 1.77 (0.85) with a linear interaction and 3.90 (1.14) with spline controls. These values are close to the biases of the same estimators in the plasmode design without any effect (0.59 and 1.05), so they are what restrictive control functions produce in these data rather than evidence of a gradient.

Years since take-off

The J-curve prediction receives no statistical support. In the baseline, the contrast between the last and the first tercile of time since take-off is -0.11 (11.17) and the constancy test does not reject ($p = 0.09$); the first tercile contains the years long before take-off in countries that adopted late, where the estimate is very imprecise. With the predetermined moderator, which is zero before take-off, the contrast is 4.95 (3.02, $p = 0.19$), and in the complete-ICT sample it is 5.72 (4.88). The more negative association in the first years after take-off that appears in the complete-ICT sample is therefore not robust to the choice of sample.

Dose response

The marginal association $\hat{f}'(d)$ is negative over the range of adoption, most clearly at low adoption (Figure 3a), and linearity of f is not rejected ($p = 0.14$). We do not detect departures from a linear conditional dose response over the evaluated support, but the linearity test has little power against curvature of the size considered in the plasmode (Section 4.3), so the absence of visible network effects or saturation is weak evidence.

Multiple testing

The three primary constancy tests give $p = 0.96$, 0.16 and 0.09, with Holm-adjusted p -values of 0.96, 0.33 and 0.28. Without the penalty, the tests give $p = 0.57$, 0.04 and 0.03 (Holm 0.57, 0.10, 0.10); the plasmode shows the unpenalised tests to have close to nominal size but low power.

5.4 Data sources and vintages

The PWT panels tell a different story about heterogeneity (Panel B of Table 4 and Figure 4b). With the same estimator, the contrast along initial productivity is 14.42 (4.92) in PWT 10.0 (1996–2019) and 14.68 (3.88) in PWT 11.0 (1996–2023), and the

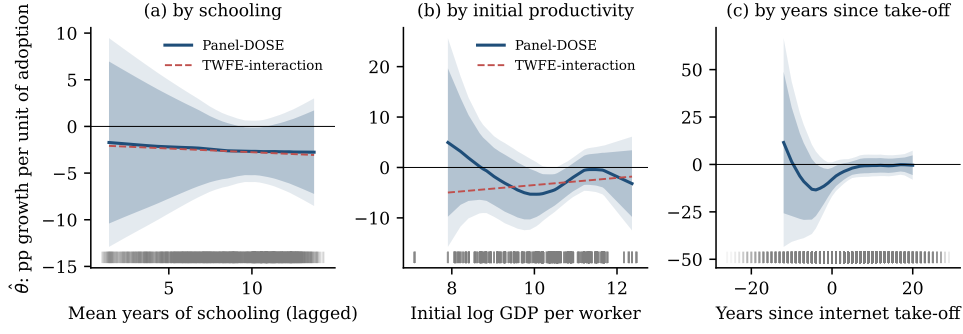


Fig. 2 Estimated association of lagged internet diffusion with labour-productivity growth, WDI panel 1996–2025: (a) by mean years of schooling and (b) by initial log GDP per worker, each with the TWFE linear-interaction estimate (dashed); (c) by years since the internet-user share first reached 10%. Dark bands: 95% pointwise; light bands: 95% simultaneous, both for the penalised fit (Remark 1). Grids span the 1st to 99th percentiles and the vertical axes show the full bands; tick marks show the distribution of the moderator.

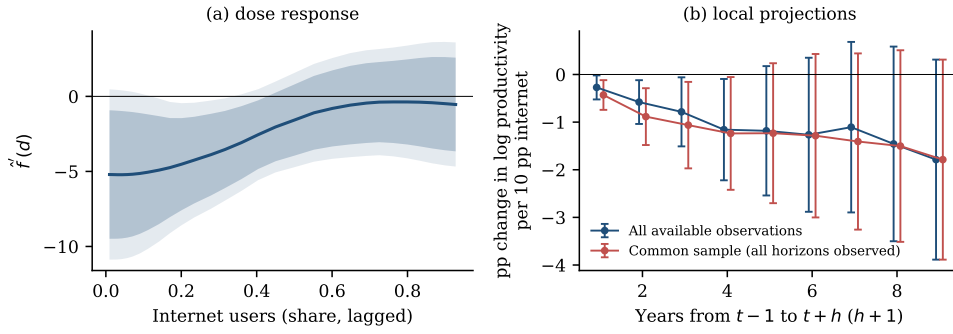


Fig. 3 (a) Marginal association $\hat{f}'(d)$ in the dose-response model, from the 13th to the 95th percentile of adoption, with 95% pointwise and simultaneous bands. (b) Local projections: change in log labour productivity ($\times 100$) from $t - 1$ to $t + h$ associated with a ten-point higher internet share in $t - 1$, plotted against the number of years covered ($h + 1$), for all available observations and for the common sample in which the nine-year change is observed, with 95% intervals.

constancy test rejects in both ($p = 0.008$ and $p < 0.001$), against 2.01 (3.55, $p = 0.61$) in the WDI data for the countries and years of the PWT 11.0 panel. The PWT gradient comes from a strongly negative association in the least productive economies. It is smaller when initial productivity is measured at national prices rather than at current PPPs (12.21 in PWT 10.0 and 8.21 in PWT 11.0), and in PWT 11.0 restricted to 1996–2019 (7.73, s.e. 4.88). The linear benchmarks also find larger gradients in PWT than in WDI (TWFE with spline controls: 6.93 in PWT 11.0 against 3.90 in WDI), but the plasmode shows that they produce gradients of this size without any true effect.

These comparisons differ in more than the productivity series: the PWT panels include countries that are not in the WDI panel, and their controls include the PWT human-capital index and the labour share instead of the dependency ratio and the urban share. Panel A of Table 5 therefore compares the two sources on the 2,839 country-years (108 countries) that both contain, replacing one component at a time and refitting each of the eight combinations. With every component from WDI the contrast is 1.89 (3.57); with every component from PWT 11.0 it is 9.29 (3.71). Because the components interact, we attribute the change with Shapley values, which average each component’s marginal contribution over all orders of replacement: the growth series accounts for 51% of the change, the control set for 38% and the moderator for 10%. The two sources agree on the ranking of countries (rank correlation of initial productivity 0.948), but their annual growth rates correlate at only 0.81, and at 0.75 after removing country and year effects—the variation the estimators use—with the weakest agreement in the least productive tercile (0.68 against 0.73 in the most productive; Table C6 in the supplementary material). The difference between the sources is stable across split seeds (from -8.58 to -7.57) and to dropping any single country, and holding the smoother fixed gives similar values (1.82 and 9.98). It is, however, imprecise: a paired country bootstrap that refits both sources on the same resampled countries gives a standard error of 4.61 and a 95% interval of $[-15.44, 1.65]$. The data therefore suggest, but do not establish at conventional levels, that heterogeneity estimates differ between the sources, and about half of the difference comes from the growth series.

Panel B makes the same comparison for the two PWT releases on the 2,655 country-years (115 countries) they have in common. The gradient is large in both: 15.09 (4.91) with every component from PWT 10.0 and 12.90 (5.55) with every component from PWT 11.0. The difference of 2.19 has a bootstrap standard error of 4.94, varies between 1.28 and 6.69 across split seeds and between -0.29 and 5.55 when single countries are dropped, and its Shapley attribution changes sign between components. The two releases rank countries almost identically (rank correlation 0.996), and their growth rates correlate at 0.89. We therefore find no evidence that the revision from PWT 10.0 to PWT 11.0 changed the gradient; the revision matters much less for our question than the choice between PWT and World Bank data.

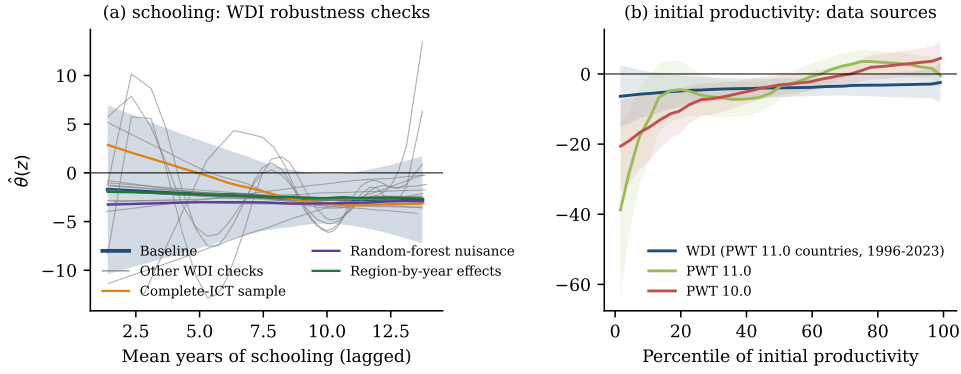


Fig. 4 (a) $\hat{\theta}(z)$ by schooling across the WDI robustness checks that keep the baseline moderator, treatment and outcome (listed in Tables 4 and C11); three checks are highlighted and the shaded band is the 95% pointwise band of the baseline. (b) $\hat{\theta}(z)$ by initial productivity in the WDI baseline, PWT 11.0 (1996–2023) and PWT 10.0 (1996–2019), with 95% pointwise bands; the horizontal axis is the percentile of the moderator within each sample, because the levels are measured in different units (constant 2021 PPP dollars in WDI, current PPPs in 2021 and 2017 US dollars in PWT 11.0 and 10.0).

Table 4 Heterogeneous associations of internet diffusion with labour-productivity growth.

Specification	Average	Tercile of moderator			High –	BLP	Constancy
	(FE-DML)	Low	Middle	High	low	slope	test p
<i>Panel A. Moderator: mean years of schooling</i>							
Baseline	–2.64** (1.33)	–2.15 (3.00)	–2.63* (1.44)	–2.72* (1.65)	–0.70 (3.60)	–0.10 (0.49)	0.96
Complete-ICT sample	–2.44** (1.22)	0.21 (2.63)	–2.75** (1.28)	–3.25* (1.68)	–3.55 (3.36)	–0.51 (0.46)	0.40
Country-specific linear trends [†]	–3.00** (1.38)	–5.96** (3.03)	–1.62 (2.01)	–0.66 (1.67)	4.99 (3.50)	0.46 (0.47)	0.22
Excluding transition economies [†]	–1.56 (1.29)	–3.07 (2.86)	–1.83 (1.53)	–0.68 (1.34)	2.22 (3.32)	0.30 (0.45)	0.47
Period 1996–2011 [†]	–4.91** (2.18)	5.99 (4.87)	–2.42 (3.03)	–6.73*** (2.39)	–12.61** (5.75)	–1.76** (0.79)	0.06
Period 2012–2025 [†]	–0.70 (2.15)	–5.06 (3.70)	0.54 (1.88)	4.19 (2.68)	9.29* (5.08)	1.29* (0.70)	0.09
Adding mobile and broadband [†]	–0.50 (1.24)	–1.40 (2.41)	–0.60 (1.24)	0.10 (1.60)	1.62 (3.02)	0.22 (0.41)	0.54
PWT 11.0, 1996–2023	–2.60* (1.33)	–11.56* (6.07)	–1.58 (1.86)	–0.82 (1.24)	11.14* (6.19)	9.64* (5.02)	0.22
PWT 10.0, 1996–2019	–2.12 (1.54)	–10.32* (5.69)	–0.95 (2.13)	–2.12* (1.09)	8.00 (5.87)	9.10** (4.49)	0.04
<i>Panel B. Moderator: initial log GDP per worker</i>							
Baseline	–2.64** (1.33)	–0.59 (3.00)	–4.79*** (1.81)	–1.27 (1.43)	–0.76 (3.27)	–0.81 (1.40)	0.16
Complete-ICT sample	–2.44** (1.22)	–1.69 (2.71)	–2.35* (1.30)	–3.05** (1.34)	–1.10 (3.18)	–0.44 (1.27)	0.68
Country-specific linear trends [†]	–3.00** (1.38)	–4.79 (3.00)	–3.23** (1.48)	–1.73 (1.48)	2.85 (3.48)	1.14 (1.39)	0.42
Excluding transition economies [†]	–1.56 (1.29)	5.16* (2.96)	–4.88** (2.35)	0.31 (1.46)	–4.83 (3.18)	–3.31*** (1.27)	0.06
Period 1996–2011 [†]	–4.91** (2.18)	–12.98 (8.07)	–7.89** (3.66)	–1.19 (1.88)	13.10 (8.05)	3.51 (3.55)	0.02
Period 2012–2025 [†]	–0.70 (2.15)	–1.54 (2.97)	–0.54 (2.17)	0.27 (3.02)	1.99 (4.32)	0.79 (1.71)	0.54
WDI data, PWT 11.0 countries, 1996–2023	–3.80*** (1.30)	–5.17* (2.69)	–3.95*** (1.32)	–3.12* (1.79)	2.01 (3.55)	0.88 (1.56)	0.61
PWT 11.0, 1996–2023	–2.60* (1.33)	–12.09*** (3.52)	–3.66** (1.62)	2.46* (1.42)	14.68*** (3.88)	7.26*** (2.07)	<0.01
PWT 10.0, 1996–2019	–2.12 (1.54)	–12.19*** (3.83)	–3.28** (1.49)	2.14 (1.83)	14.42*** (4.92)	6.50*** (2.22)	0.01
<i>Panel C. Moderator: years since internet take-off (share ≥ 10%)</i>							
Baseline	–1.94 (1.38)	–0.26 (11.14)	–3.18* (1.67)	–0.74 (1.63)	–0.11 (11.17)	–0.28 (0.57)	0.09
Complete-ICT sample	–1.86 (1.31)	–6.41 (4.19)	–2.54 (1.59)	–0.73 (1.72)	5.72 (4.88)	0.24 (0.22)	0.49
Predetermined moderator (zero before take-off) [†]	–1.75 (1.38)	–4.62** (2.34)	–2.14 (1.54)	0.05 (1.76)	4.95 (3.02)	0.25 (0.18)	0.19

Notes: Panel-DOSE estimates; standard errors clustered by country in parentheses. Coefficients are per unit of adoption (0–1); divide by 10 for a ten-point difference. The dependent variable is annual labour-productivity growth (%) unless stated otherwise. Baseline: WDI panel (149 countries, 1996–2025) with the controls listed in Section 5. “Complete-ICT sample”: observations with every ICT series, the investment share, lagged productivity and the second lag of the treatment observed (141 countries). “Country-specific linear trends” removes a linear trend for each country in the within step. “Excluding transition economies” drops the post-socialist economies of Eastern Europe and Central Asia. PWT rows use the Penn World Table panels, with the PWT human-capital index (Panel A) and initial log PWT labour productivity at current PPPs (Panel B) as moderators. The BLP slope is per unit of the moderator (years of schooling, log points, years). “Constancy test p ”: sup- t test that θ is constant. [†]: ten repetitions of the sample split (20 otherwise), one penalty pooled over the splits. The high-minus-low contrast is aggregated across splits as a functional in its own right, so it can differ slightly from the difference of the two tercile columns. *, **, ***: 10%, 5%, 1%.

Table 5 Heterogeneity along initial productivity across data sources, on common country-years: component replacement.

Source of			Refitted			Fixed smoother	
Outcome	Moderator	Controls	High – low	(s.e.)	$\hat{\lambda}$	High – low	(s.e.)
<i>Panel A. WDI and PWT 11.0 (2,839 country-years, 108 countries, 1996–2023)</i>							
WDI	WDI	WDI	1.89	(3.57)	10^4	1.82	(3.26)
WDI	WDI	PWT 11.0	3.63	(3.52)	10^4	3.87	(3.25)
WDI	PWT 11.0	WDI	1.91	(3.42)	10^4	2.37	(3.42)
WDI	PWT 11.0	PWT 11.0	4.51	(3.36)	10^4	4.86	(3.54)
PWT 11.0	WDI	WDI	4.63	(3.02)	10^4	4.57	(3.10)
PWT 11.0	WDI	PWT 11.0	7.86**	(3.40)	17.8	8.18**	(3.25)
PWT 11.0	PWT 11.0	WDI	5.47	(3.47)	10^4	5.83*	(3.37)
PWT 11.0	PWT 11.0	PWT 11.0	9.29**	(3.71)	10^4	9.98***	(3.61)
<i>Shapley attribution of the change (PWT 11.0 – WDI)</i>							
Outcome			3.81			3.92	
Moderator			0.77			1.16	
Controls			2.82			3.08	
Total			7.40			8.16	
Difference WDI – PWT 11.0: –7.40; paired bootstrap s.e. 4.61, 95% interval [–15.44, 1.65]							
Across 10 split seeds: difference from –8.58 to –7.57; leave-one-country-out: –11.04 to –6.76							
<i>Panel B. PWT 10.0 and PWT 11.0 (2,655 country-years, 115 countries, 1996–2019)</i>							
PWT 10.0	PWT 10.0	PWT 10.0	15.09***	(4.91)	10^3	17.10***	(5.51)
PWT 10.0	PWT 10.0	PWT 11.0	12.78***	(4.95)	1,778	15.24***	(5.59)
PWT 10.0	PWT 11.0	PWT 10.0	15.74***	(5.47)	0.562	15.20***	(5.35)
PWT 10.0	PWT 11.0	PWT 11.0	14.94***	(5.43)	0.562	13.69**	(5.54)
PWT 11.0	PWT 10.0	PWT 10.0	12.70**	(5.35)	10^2	14.09***	(5.37)
PWT 11.0	PWT 10.0	PWT 11.0	8.91	(5.44)	10^4	11.14*	(5.93)
PWT 11.0	PWT 11.0	PWT 10.0	13.73***	(5.18)	1	12.82**	(5.45)
PWT 11.0	PWT 11.0	PWT 11.0	12.90**	(5.55)	0.562	11.48*	(5.97)
<i>Shapley attribution of the change (PWT 11.0 – PWT 10.0)</i>							
Outcome			–2.45			–2.82	
Moderator			2.08			–0.99	
Controls			–1.81			–1.81	
Total			–2.19			–5.61	
Difference PWT 10.0 – PWT 11.0: 2.19; paired bootstrap s.e. 4.94, 95% interval [–10.36, 10.57]							
Across 10 split seeds: difference from 1.28 to 6.69; leave-one-country-out: –0.29 to 5.55							

Notes: Panel-DOSE tercile contrast along initial log labour productivity on the country-years present in both sources; the treatment is the same ITU series in all rows. Each row takes the outcome, the moderator and the control set from the source indicated (WDI controls: schooling, government share, openness, population growth, dependency ratio, urban share; PWT controls: human-capital index, government share, openness, population growth, labour share). “Refitted”: nuisance functions, cross-validated penalty and tercile groups re-estimated for each configuration, 20 splits; standard errors conditional on the selected penalty. “Fixed smoother”: moderator transformed to within-source percentile ranks, tercile groups of the second source and one penalty for every row. Shapley values average the marginal contribution of each component over the six orders of replacement and sum to the total change. Paired country bootstrap: 199 draws of countries with replacement, both sources refitted on each draw (ten splits, copies of a country kept in the same fold); basic (bias-corrected) percentile interval. Split seeds: ten seeds of the two all-component fits; leave-one-country-out: both sources refitted without each country, with penalty, seed and tercile cutoffs fixed. *, **, ***: 10%, 5%, 1%.

6 Discussion

6.1 Interpretation

Three features of the results deserve emphasis. First, within countries, years with a higher lagged internet-user share were not years of faster productivity growth. The linear estimate is negative and stable across trends, convergence controls and development-specific year effects; the flexible estimate is negative in almost every specification but its interval includes zero with two-way clustering, for some split seeds and in many robustness checks. We therefore do not claim a significant negative association. What the data rule out, under our assumptions, is a positive within-country association of the size reported by earlier macro studies. The association is concentrated in 1996–2011, is weaker without the transition economies, and is shared with the co-diffusing mobile and broadband series. It is a low-frequency co-movement between the level of adoption and growth, and changes in adoption are not associated with growth. The literature suggests several explanations, which our design cannot distinguish: the J-curve argument that intangible investment and reorganisation depress measured productivity during diffusion ([Brynjolfsson et al. 2021](#)), the concentration of ICT returns in firms with complementary capabilities ([Gal et al. 2019](#); [Nicoletti et al. 2020](#)), confounding by the growth slowdown that followed the 2000s boom in economies that adopted late, and other unobserved confounding. A shift of employment towards low-productivity activities does not fit the decomposition, because the association is with output growth rather than with employment growth. A non-positive within-country association is also consistent with a positive cross-country association in long differences, which reflects which economies adopted, not what adoption did within them.

Second, none of the heterogeneity predictions is supported robustly. With our precision, we cannot distinguish a flat function from moderate gradients of either sign, so the absence of evidence for complementarities with schooling is not evidence of their absence. Along initial productivity, some flexible fits suggest non-monotone variation, but the evidence is weak and not of the form the theories predict. The J-curve pattern along time since take-off that appears in the complete-ICT sample does not survive in the full sample.

Third, estimates of where the association differs are sensitive to the data. The gradient along initial productivity is large in both PWT releases and much weaker in WDI data for the same country-years. On identical country-years, about half of the difference is due to the growth series, whose within-country agreement between the two sources is weakest in the least productive economies, and a third to the control set; the difference is stable across split seeds but imprecise. The revision between PWT releases, by contrast, does not change the gradient detectably. These findings extend the evidence of [Johnson et al. \(2013\)](#) and [Ponomareva and Katayama \(2010\)](#) on data choices to heterogeneous associations of internet diffusion, and they are a warning against reading such gradients as stable facts. The linear benchmarks point the other way in the WDI data: TWFE with a linear interaction gives a tercile contrast of 1.77 (0.85) along initial productivity and TWFE with spline controls 3.90 (1.14). In the plasmode design without any effect, the same estimators produce gradients of similar

size and reject in most replications (Table C9), so we do not take this as evidence of a gradient.

6.2 Policy implications

Because the estimates are conditional associations, their policy content is limited, and they do not identify the return to connectivity investment. With that qualification, they caution against justifying connectivity programmes by expected short-run gains in aggregate measured productivity: in the economies and period we study, years with higher internet use were not followed by faster measured productivity growth within countries over horizons of up to nine years. This does not imply that such programmes are undesirable. Connectivity has consumer-surplus, inclusion and service-delivery benefits that productivity statistics do not capture (World Bank 2016; Goldfarb and Tucker 2019; Hjort and Tian 2025), and averages across countries can hide large gains for specific groups within them. The literature offers several explanations for a missing aggregate pay-off that our data cannot test: gains may depend on complementary investment in firm capabilities, management and skills (Bloom et al. 2012; Gal et al. 2019), on the intensity and quality of use rather than on access (Comin and Mestieri 2018), or on reorganisation that takes longer than our horizons (Brynjolfsson et al. 2021). These are hypotheses for further work, not implications of our estimates. Finally, cross-country benchmarks that claim to identify the stage of development at which digital investment pays off should be treated with caution, because such heterogeneity estimates change with data releases. Policy-relevant causal statements would require designs with credible exogenous variation, such as the staggered arrival of infrastructure used by Hjort and Poulsen (2019).

6.3 Limitations

The estimates are conditional associations of a lagged adoption level with growth, and the assumptions behind them are strong. Strict exogeneity rules out feedback from productivity shocks to future adoption; the lead test does not reject it but has little power, lagging the treatment by one or two years mitigates but does not eliminate reverse causality, and the instrument we could construct from historical fixed-line density is weak. Nuisance sufficiency requires that the controls and their country means capture the confounding path, which rules out, for example, global shocks with country-specific loadings that are correlated with adoption, or development trends that the controls do not track; region-by-year effects address the first concern only partly. The baseline avoids the mechanical violation of strict exogeneity that lagged productivity controls create, but other controls, such as urbanisation, may respond to past productivity, and convergence dynamics are absorbed only to the extent that initial productivity and the country means capture them. Measurement error matters in both directions: employment in low-income economies is largely modelled by the ILO and ITU internet shares for these economies are often estimated. In a linear model classical error in adoption would attenuate the coefficient; in our residualised varying-coefficient model the direction of the bias is not known in general. Results are similar

with GDP-per-capita growth as the outcome, in countries with above-median statistical capacity, and without screening the adoption series, but they cannot rule out non-classical errors correlated with development. The last observations rest on provisional national accounts and schooling in 2024 is extrapolated. Internet penetration measures access, not intensity or quality of use; for low-income economies, mobile-network coverage and the quality of connections (3G or 4G) may matter more than the share of users, and ITU user shares for many of them are estimated from subscriptions rather than measured in household surveys. For economies without labour-force surveys, the ILO models employment with predictors that include demographic and urbanisation variables, two of which are among our controls; errors in the denominator of productivity can therefore be correlated with the controls. Productivity statistics also miss the value of free digital goods, and aggregate data cannot reveal effects that differ by gender, between rural and urban areas, or between formal and informal firms. Coverage of low-income and conflict-affected economies is incomplete (14 of 25 low-income economies are included), so the policy caution applies mainly to middle- and high-income economies. Finally, the asymptotic theory covers the fixed- K sieve with a vanishing penalty and, for the shape tests, a fixed penalty (Propositions 3 and 4); the data-dependent penalty is evaluated by simulation only, and the bands describe the penalised fit rather than θ itself (Remark 1).

7 Conclusion

Over three decades of internet diffusion in 149 economies, we do not find a positive within-country association between lagged internet use and labour-productivity growth. The linear estimate is negative and robust to trends and convergence controls; the flexible estimate is negative but not clearly different from zero, and the association is concentrated in the first half of the period. We find no robust variation of the association with schooling, initial productivity or time since take-off, within the precision of the data. Heterogeneity along the development dimension appears in the Penn World Table and is much weaker in World Bank data for the same country-years, largely because the growth series differ, while the revision between PWT releases does not change it detectably. These are conditional associations, not causal effects of connectivity.

Methodologically, Panel-DOSE combines orthogonal series estimation, Mundlak features, country-level cross-fitting, a within transformation and an O’Sullivan penalty, whose null space makes the constancy and linearity tests valid for a fixed penalty. In a plasmode design built on our data, the constancy test with the cross-validated penalty has size between 0.06 and 0.09 at the 5% level; finite-sample calibration of this kind should accompany applications, because the theory does not cover the data-dependent penalty. In our simulations it is most useful when confounding is strongly nonlinear; when it is mild, TWFE with spline controls does as well. Most of its advantage comes from the Mundlak features, and linear interaction models give unreliable intervals when the control function is misspecified. The code, which also reports identifying variation, paired-bootstrap comparisons of estimates and shape tests, is provided as an installable Python module in the replication package.

Future work could combine the estimator with instruments or staggered infrastructure roll-outs to move from associations to effects, use firm- or region-level measures of the intensity of digital use, and examine the moderating roles of institutions, competition and data governance.

Declarations

Data and code availability. The data are publicly available from the World Bank’s World Development Indicators (July 2026 release), the UNDP Human Development Report 2025 and the Penn World Table (versions 11.0 and 10.0). The WDI snapshot used here was retrieved through the World Bank API in September 2026 and is archived with SHA-256 checksums in the replication package, which reads the archived snapshot by default and verifies it before the panels are built. The package contains the Python implementation of Panel-DOSE, the scripts that reproduce all tables, figures and numbers in the text, and the list of package versions used. The package is available at <https://github.com/vinhqdang/Journal-of-Digital-Economy>; an archived version with a persistent identifier will be deposited on acceptance.

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Use of generative AI. During the preparation of this work the author(s) used generative AI tools to assist with writing and debugging code and with drafting and editing the text. The research idea, the model and the algorithm are the author(s)’ own. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Appendix A Proofs

Throughout, the year effect is a function of the time index t , which is an element of F_{it} , so λ_t is absorbed by ℓ . The two-way operator $\mathcal{M}$ is the orthogonal projection onto the complement of the space spanned by country and year indicators; it is symmetric and idempotent, and $\mathcal{M}c = 0$ for any vector with elements $c_{it} = c_i + c_t$. For country i , collect its T_i observations into Y_i , Φ_i ($T_i \times K$) and F_i , and write $V_i = \Phi_i - m_0(F_i)$ and $e_i = Y_i - \ell_0(F_i) - V_i\beta_0$. Under Assumptions 1–2 and $r_K = 0$, $e_{it} = a_i + \varepsilon_{it}$. Let $Q_i = I_{T_i} - T_i^{-1}\iota\iota'$ be the country-demeaning matrix, $Q = \text{diag}(Q_1, \dots, Q_G)$, and let A be the $n \times \mathcal{T}$ matrix of indicators of the $\mathcal{T}$ calendar years, with country blocks A_i . By the Frisch–Waugh–Lovell theorem the two-way operator is, for balanced and

unbalanced panels alike,

$$\mathcal{M} = Q - QA(A'QA)^+A'Q,$$

where $^+$ denotes the Moore–Penrose inverse ($A'QA$ has rank $\mathcal{T} - 1$ because A_t lies in the span of the country indicators). The alternating-projection algorithm used in the code converges to this orthogonal projection (the von Neumann–Halperin theorem; see [Gaure 2013](#)). The following lemma allows us to replace $\mathcal{M}$ by Q in the asymptotic arguments.

Assumption 5 (Panel structure) (a) The number of calendar years $\mathcal{T}$ is fixed. (b) The set of years in which country i is observed, $\mathcal{S}_i$, is independent across countries, and $\mathbb{E}[\varepsilon_{it} | \mathcal{S}_i, D_i, X_i, \alpha_i] = 0$ and $\mathbb{E}[D_{it} | \mathcal{S}_i, F_{i1}, \dots, F_{iT_i}] = \mathbb{E}[D_{it} | F_{it}]$, i.e. the unbalancedness is ignorable given the observed paths. (c) The smallest non-zero eigenvalue of $G^{-1}A'QA$ is bounded away from zero with probability approaching one; this holds, for example, if each calendar year is observed, together with at least one other year, for a fixed positive fraction of the countries.

Lemma 1 (Year demeaning) Let W and E be n -vectors (or $n \times K$ arrays) with country blocks W_i, E_i that are independent across countries, have bounded fourth moments, and satisfy $\mathbb{E}[A'_i Q_i W_i] = 0$ and $\mathbb{E}[A'_i Q_i E_i] = 0$. Under Assumption 5, $G^{-1/2}W'ME = G^{-1/2} \sum_i W'_i Q_i E_i + O_p(G^{-1/2})$ and $G^{-1}W'ME = G^{-1} \sum_i W'_i Q_i E_i + O_p(G^{-1})$.

Proof By the expression for $\mathcal{M}$,

$$G^{-1/2}W'ME = G^{-1/2}W'QE - (G^{-1/2}W'QA)(G^{-1}A'QA)^+(G^{-1/2}A'QE)G^{-1/2}.$$

The vector $G^{-1/2}A'QE = G^{-1/2} \sum_i A'_i Q_i E_i$ is a normalised sum of G independent mean-zero terms with bounded variance, hence $O_p(1)$ by Chebyshev's inequality, and the same holds for $G^{-1/2}A'QW$. By Assumption 5(c), $\|(G^{-1}A'QA)^+\| = O_p(1)$; because the Moore–Penrose inverse is applied on the row space of $A'Q$, in which both $A'QW$ and $A'QE$ lie, no further condition is needed for the rank-deficient direction. The second term is therefore $O_p(1) \cdot O_p(1) \cdot O_p(1) \cdot G^{-1/2} = O_p(G^{-1/2})$. The second statement follows in the same way. $\square$

In the applications of the lemma, W is $V = \Phi - m_0(F)$ or a nuisance error, and E is e or a nuisance error. The condition $\mathbb{E}[A'_i Q_i V_i] = 0$ holds because $Q_i A_i$ is a function of the observed years, which are coordinates of F , and $\mathbb{E}[V_{it} | \mathcal{S}_i, F_{i1}, \dots, F_{iT_i}] = 0$ by Assumptions 3 and 5(b); $\mathbb{E}[A'_i Q_i e_i] = \mathbb{E}[A'_i Q_i \varepsilon_i] = 0$ by Assumptions 1 and 5(b). For the nuisance errors, which are estimated on other countries and are therefore fixed functions conditionally on the training folds, the same algebra shows that the correction term is bounded by $O_p(1)$ times the product of their L_2 norms and $G^{-1/2}\|A'QV\|$ or $G^{-1/2}\|A'Qe\|$, which is of smaller order than the terms already controlled in the proof of Proposition 3.

Proof of Proposition 1

(i) For directions $\delta_\ell(F)$ and $\delta_m(F)$, and using Lemma 1,

$$\psi(\beta_0; \ell_0 + r\delta_\ell, m_0 + r\delta_m) = \mathbb{E}[(V_i - r\delta_{m,i})' Q_i (e_i - r\delta_{\ell,i} + r\delta_{m,i}\beta_0)].$$

Differentiating at $r = 0$ gives $-\mathbb{E}[\delta'_{m,i} Q_i e_i] + \mathbb{E}[V_i' Q_i (\delta_{m,i}\beta_0 - \delta_{\ell,i})]$. In the first term, $Q_i e_i = Q_i \varepsilon_i$ because a_i is constant within country, and $\mathbb{E}[\varepsilon_{it} | F_{i1}, \dots, F_{iT_i}] = 0$ by Assumption 1, because F_{is} is a function of (D_i, X_i) ; the term is zero. In the second term, $Q_i w_i$ is a function of $(F_{i1}, \dots, F_{iT_i})$ for $w_i = \delta_{m,i}\beta_0 - \delta_{\ell,i}$, and, when Z_{it} is in F_{it} , $\mathbb{E}[V_{it} | F_{i1}, \dots, F_{iT_i}] = b(Z_{it})\{\mathbb{E}[D_{it} | F_{i1}, \dots, F_{iT_i}] - \mu(F_{it})\} = 0$ by Assumption 3 (in the dose-response model the same condition is imposed on Φ); the term is zero.

(ii) $\mathcal{M}(Y - \ell - c) = \mathcal{M}(Y - \ell) - \mathcal{M}c = \mathcal{M}(Y - \ell)$ whenever $c_{it} = c_i + c_t$, because $\mathcal{M}$ annihilates the span of the country and year indicators; the same holds for each column of m . If instead $\hat{\mu} = \mu + c_i$, the regressors become $\mathcal{M}\{b(Z)(D - \mu)\} - \mathcal{M}\{b(Z_{it})c_i\}$. If $b(Z_{it}) = \bar{b}_i$ does not vary within country (in particular if $K = 1$, where $b \equiv 1$, or if Z is time-invariant), then $b(Z_{it})c_i = \bar{b}_i c_i$ is a country-specific constant vector and $\mathcal{M}\{b(Z_{it})c_i\} = 0$. When $b(Z_{it})$ varies within country the term is in general non-zero. Note that multiplication by c_i does not commute with the two-way operator, $\mathcal{M}\{c_i b(Z_{it})\} \neq c_i \mathcal{M}\{b(Z_{it})\}$ in general, because $\mathcal{M}$ also removes year means; for example, with two countries observed in three years, $c = (1, 1, 1, 3, 3, 3)'$ and $b = (0, 1, 4, 1, 3, 8)'$, $\mathcal{M}(cb) = (3.67, 1.17, -4.83, -3.67, -1.17, 4.83)'$ whereas $c \odot \mathcal{M}b = (0.67, 0.17, -0.83, -2.00, -0.50, 2.50)'$. $\square$

Proof of Proposition 2

If Z_{it} is a coordinate of F_{it} , then $b_k(Z_{it})$ is F_{it} -measurable and $\mathbb{E}[D_{it} b_k(Z_{it}) | F_{it}] = b_k(Z_{it})\mathbb{E}[D_{it} | F_{it}]$. Consequently $\Phi_{it} - m(F_{it}) = b(Z_{it})\{D_{it} - \mu(F_{it})\}$. $\square$

Proof of Proposition 3

Write $\tilde{\Phi} = \mathcal{M}(\Phi - \hat{m})$, $\tilde{Y} = \mathcal{M}(Y - \hat{\ell})$, and let $\tilde{\Phi}^0$ be the same object at the true nuisance functions. From (3),

$$\sqrt{G}(\hat{\beta}_\lambda - \beta_K) = \left(\frac{1}{G} \tilde{\Phi}' \tilde{\Phi} + \lambda \frac{\kappa}{G} \Omega \right)^{-1} \left\{ \frac{1}{\sqrt{G}} \tilde{\Phi}' (\tilde{Y} - \tilde{\Phi} \beta_K) - \sqrt{G} \lambda \frac{\kappa}{G} \Omega \beta_K \right\}.$$

The last term vanishes because $\kappa/G = O_p(1)$ and $\lambda_G \sqrt{G} \rightarrow 0$. Let $\Delta_m = \hat{m} - m_0$ and $\Delta_\ell = \hat{\ell} - \ell_0$ on the evaluation fold. By Lemma 1,

$$\frac{1}{\sqrt{G}} \tilde{\Phi}' (\tilde{Y} - \tilde{\Phi} \beta_K) = \frac{1}{\sqrt{G}} \sum_i V_i' Q_i e_i + R_1 + R_2 + R_3 + O_p(G^{-1/2}),$$

where $R_1 = -G^{-1/2} \sum_i \Delta'_{m,i} Q_i e_i$, $R_2 = G^{-1/2} \sum_i V_i' Q_i (\Delta_{m,i} \beta_K - \Delta_{\ell,i})$ and $R_3 = -G^{-1/2} \sum_i \Delta'_{m,i} Q_i (\Delta_{m,i} \beta_K - \Delta_{\ell,i})$. Because the nuisance functions are estimated on countries outside the evaluation fold, conditionally on the training sample the summands of R_1 and R_2 are independent across countries with mean zero (by Assumptions 1 and 3, as in the proof of Proposition 1), and their conditional variances are

bounded by a constant times $\|\Delta_m\|_2^2$ and $\|\Delta_m\|_2^2 + \|Q\Delta_\ell\|_2^2$; hence $R_1, R_2 = o_p(1)$. By the Cauchy–Schwarz inequality, $\|R_3\| \leq C\sqrt{G}\|\Delta_m\|_2(\|\Delta_m\|_2 + \|Q\Delta_\ell\|_2)$, which is $o_p(1)$ under Assumption 4(b); by Proposition 2, $\|\Delta_m\|_2 \leq C\|\hat{\mu} - \mu\|_2$. Only the country-demeaned error $Q\Delta_\ell$ appears because Q annihilates country constants. Similarly $G^{-1}\tilde{\Phi}'\tilde{\Phi} \xrightarrow{p} J$. The Lindeberg–Feller central limit theorem for independent clusters gives $G^{-1/2}\sum_i V_i'Q_i e_i = G^{-1/2}\sum_i s_i \xrightarrow{d} N(0, \Omega_s)$. Consistency of $G\hat{V}$ follows from standard arguments for cluster-robust variance estimators (Cameron and Miller 2015). For a fixed number S of splits, each $w'\hat{\beta}_s$ has the same linear representation, so the median does too and the variance adjustment is asymptotically negligible (Chernozhukov et al. 2018). $\square$

Proof of Proposition 4

Let $A_\lambda = \tilde{\Phi}'\tilde{\Phi} + \lambda\kappa\Omega$. From (3), $\hat{\beta}_\lambda = A_\lambda^{-1}\tilde{\Phi}'\tilde{Y}$, so that $\hat{\beta}_\lambda - \tilde{\beta}_\lambda = A_\lambda^{-1}\tilde{\Phi}'(\tilde{Y} - \tilde{\Phi}\beta_K)$. The expansion of $G^{-1/2}\tilde{\Phi}'(\tilde{Y} - \tilde{\Phi}\beta_K)$ in the proof of Proposition 3 does not involve the penalty, and $G^{-1}A_\lambda \xrightarrow{p} J_\lambda$, which is positive definite because J is. The first claim follows. If $\theta(z) = a + bz$, then θ lies in the span of the cubic B-spline basis, $r_K = 0$, and $b(z)'\beta_K$ has zero second derivative, so $\beta_K'\Omega\beta_K = 0$; because Ω is positive semi-definite, $\Omega\beta_K = 0$ and $\tilde{\beta}_\lambda = A_\lambda^{-1}(A_\lambda - \lambda\kappa\Omega)\beta_K = \beta_K$. The residuals $\hat{u}$ then differ from the true within-transformed errors only by terms controlled in the proof of Proposition 3, so $G\hat{V}$ is consistent. The linearity statistic is a continuous function of $\sqrt{G}ML_g(\hat{\beta}_\lambda - \beta_K)$, where L_g evaluates $b(\cdot)$ on the grid and M removes the linear fit on the grid, and $ML_g\beta_K = 0$ under the null; its limit distribution is that of the maximum of a Gaussian vector with the estimated covariance, which is what the simulated critical value approximates. A constant θ is a special case of a linear one. $\square$

Appendix B Data and implementation details

Variables

Main panel (1996–2025). From the World Development Indicators (July 2026 release, retrieved through the World Bank API in September 2026): GDP per person employed at constant 2021 PPP dollars (SL.GDP.PCAP.EM.KD); individuals using the internet (IT.NET.USER.ZS); mobile cellular (IT.CEL.SETS.P2) and fixed broadband (IT.NET.BBND.P2) subscriptions per 100 people; gross capital formation (NE.GDI.TOTL.ZS), government final consumption (NE.CON.GOV.T.ZS), exports (NE.EXP.GNFS.ZS) and imports (NE.IMP.GNFS.ZS) as percentages of GDP; population growth (SP.POP.GROW) and total population (SP.POP.TOTL); the urban population share (SP.URB.TOTL.IN.ZS); the age-dependency ratio (SP.POP.DPND); GDP per capita at constant 2015 dollars (NY.GDP.PCAP.KD); the Statistical Performance Indicators overall score (IQ.SPI.OVRL, country mean over 2016–2024); fuel and ores-and-metals exports as percentages of merchandise exports (TX.VAL.FUEL.ZS.UN, TX.VAL.MMTL.ZS.UN); resource exporters are countries whose combined share averages at least 50% over 1996–2025); fixed telephone subscriptions per 100 people in 1990 (IT.MLT.MAIN.P2); and learning-adjusted years of schooling (HD.HCI.LAYS, country mean). Mean years of schooling is taken from the UNDP Human Development Report

2025 composite-indices time series (1990–2023); the 2024 value is the 2023 value plus the country’s 2019–2023 linear trend. Initial log GDP per worker is the average of the available values in 1991–1995 (for Montenegro, the first available value, 1997). Percentages are divided by 100. Broadband subscriptions are set to zero in the years before a country’s first reported value, on the assumption that the service did not exist; countries that never report broadband are left missing, and a robustness check leaves all pre-report years missing. Regional and income aggregates are excluded. Region-by-year effects use the seven World Bank regions.

Screening of adoption series. Values are set to missing when the internet share is outside $[0, 100]$ or changes by more than 25 points in a year, when mobile subscriptions exceed 250 per 100 people or change by more than 100 per 100, or when broadband subscriptions exceed 60 per 100 or change by more than 20 per 100. The rules flag 39 values (8 internet, 18 mobile, 13 broadband). Only the internet flags affect the baseline sample. The list is part of the replication package. The unscreened sample, built with the same trimming cutoffs and countries as the baseline (4,081 observations), gives a headline estimate of -2.71 (1.29). The untrimmed sample keeps the baseline countries and all their observations (4,157).

PWT panels. From the Penn World Table 11.0 (1996–2023) and 10.0 (1996–2019), with identical definitions: labour productivity growth is the log change of real GDP at constant national prices (`rgdpna`) divided by persons engaged (`emp`); initial productivity is the 1991–1995 average of log output-side real GDP at current PPPs (`cgdpo`) per person engaged, and the national-prices alternative uses `rgdpna`; TFP growth is the log change of `rtfpna`; the human capital index is `hc`; the government share is `csh_g`; openness is `csh_x`–`csh_m`, which equals exports plus imports as shares of GDP because PWT records `csh_m` with a negative sign; the labour share is `labsh`. The adoption series are the ITU data distributed by Our World in Data. All controls and the treatment enter with a one-year lag in all panels.

Invariance check

We add country-specific constants to the cross-fitted residuals of the baseline and re-solve the final stage with the penalty held at its selected value (`code/invariance_check.py`). Constants added to the outcome residuals (standard deviation 1) leave every estimate unchanged up to numerical precision, as Proposition 1(ii) states. Constants added to the treatment residuals (standard deviation 0.05, against a residual standard deviation of 0.067) leave the constant-effect estimate unchanged. They also leave the Panel B estimates unchanged, because initial productivity is time-invariant, so that $b(Z_{it})$ is constant within country. For the time-varying schooling moderator they move the tercile contrast from -0.70 to -0.54 , about 0.05 of its standard error.

Sample

Table B1 shows how the baseline sample is obtained from the WDI country-years. Montenegro, which has no productivity data before 1997, is the only country whose initial productivity is not a 1991–1995 average; it takes its first available value, and excluding it leaves the results unchanged (Table C12).

Table B1 Sample construction, WDI panel.

Step	Observations	Countries
Country-years 1996-2025 (WDI economies)	6,510	217
Outcome observed	5,257	177
Treatment observed	5,042	177
Schooling observed	4,830	174
Government share observed	4,344	166
Openness observed	4,314	165
Population growth observed	4,314	165
Dependency ratio observed	4,314	165
Urban share observed	4,314	165
Initial productivity observed	4,314	165
Outcome within 1st-99th percentiles	4,226	165
Countries with at least 15 years (baseline)	4,075	149
Common sample of the earlier version	3,787	141

Notes: each row applies one additional condition to the previous row. “Observed” refers to the variable entering the regression (lagged where applicable). The last row is the complete-ICT sample used as a robustness check, which also requires mobile and broadband subscriptions, the investment share, lagged productivity and growth, and the second lag of the treatment, with trimming cutoffs computed on that sample.

Countries

Table B2 lists the 149 countries of the main panel by tercile of initial GDP per worker.

Table B2 Countries in the main estimation sample, by tercile of initial GDP per worker.

Group	Countries
Low	Albania; Armenia; Azerbaijan; Bangladesh; Benin; Bolivia; Bosnia and Herzegovina; Burkina Faso; Cabo Verde; Cambodia; Cameroon; Central African Republic; Chad; China; Comoros; Congo, Dem. Rep.; Equatorial Guinea; Gambia, The; Georgia; Ghana; Guinea; Guinea-Bissau; Haiti; Honduras; India; Indonesia; Kenya; Kyrgyz Republic; Lao PDR; Lesotho; Madagascar; Mali; Mongolia; Mozambique; Nepal; Nicaragua; Niger; Pakistan; Philippines; Rwanda; Senegal; Sierra Leone; Solomon Islands; Sri Lanka; Tajikistan; Tanzania; Timor-Leste; Togo; Tonga; Uganda; Uzbekistan; Viet Nam; Zambia
Middle	Angola; Belarus; Belize; Botswana; Brazil; Bulgaria; Chile; Colombia; Congo, Rep.; Costa Rica; Croatia; Dominican Republic; Ecuador; Egypt, Arab Rep.; El Salvador; Estonia; Fiji; Guatemala; Iraq; Kazakhstan; Korea, Rep.; Latvia; Lebanon; Lithuania; Malaysia; Mauritania; Mauritius; Moldova; Morocco; Namibia; North Macedonia; Panama; Paraguay; Peru; Poland; Romania; Sao Tome and Principe; Serbia; Slovak Republic; South Africa; Syrian Arab Republic; Thailand; Tunisia; Turkiye; Turkmenistan; Ukraine; Uruguay; West Bank and Gaza; Zimbabwe
High	Algeria; Argentina; Australia; Austria; Bahamas, The; Bahrain; Belgium; Brunei Darussalam; Canada; Cyprus; Czechia; Denmark; Finland; France; Gabon; Germany; Greece; Hong Kong SAR, China; Hungary; Iceland; Iran, Islamic Rep.; Ireland; Israel; Italy; Japan; Kuwait; Libya; Luxembourg; Malta; Mexico; Montenegro; Netherlands; New Zealand; Norway; Oman; Portugal; Qatar; Russian Federation; Saudi Arabia; Singapore; Slovenia; Spain; Sweden; Switzerland; United Arab Emirates; United Kingdom; United States

Implementation

The nuisance learners are LightGBM gradient-boosted trees (Ke et al. 2017) with 15 leaves, a minimum of 20 observations per leaf (10 in the simulations), row and column subsampling of 80% and a learning rate of 0.03; the random-forest check uses 400 trees (Breiman 2001) with a minimum leaf size of 5 and half of the features at

each split. The feature set contains the lagged controls, their country means (except for time-invariant variables), the country means of the treatment and the moderator, and the calendar year. Knots are placed at equally spaced quantiles of the moderator between its 0.5th and 99.5th percentiles, and values outside this range are assigned to the boundary. Simultaneous bands and sup- t tests use 20,000 Gaussian draws. A full fit on the empirical sample (two nuisance functions, five folds, 20 repetitions) takes about 30 seconds on one CPU core, and re-estimating the final stage with another sieve takes a fraction of a second.

Appendix C Additional results

Descriptive statistics

Table C3 Descriptive statistics, baseline sample, 1996–2025.

Variable	Obs.	Mean	Std. dev.	Min	Max
Labour-productivity growth (%)	4075	1.622	3.458	-10.758	12.459
Internet users, share ($t - 1$)	4075	0.367	0.333	0.000	1.000
Mobile subscriptions per capita ($t - 1$)	4030	0.751	0.523	0.000	2.471
Fixed broadband per capita ($t - 1$)	3995	0.091	0.128	0.000	0.489
Mean years of schooling ($t - 1$)	4075	8.480	3.270	0.975	14.325
Initial log GDP per worker (pre-sample)	4075	10.157	1.106	7.097	12.470
Investment share ($t - 1$)	4047	0.240	0.073	-0.024	0.713
Government consumption share ($t - 1$)	4075	0.161	0.070	0.020	1.477
Trade openness ($t - 1$)	4075	0.855	0.534	0.156	4.426
Population growth, % ($t - 1$)	4075	1.347	1.506	-10.927	21.700
Age-dependency ratio ($t - 1$)	4075	0.604	0.177	0.173	1.140
Urban population share ($t - 1$)	4075	0.605	0.215	0.103	1.000
Years since internet take-off ($t - 1$)	4075	4.070	10.493	-26.000	29.000
GDP per capita growth (%)	4075	2.107	3.686	-21.001	16.071

Sources: World Development Indicators (World Bank, July 2026 release), including ITU adoption series; UNDP Human Development Report 2025 (mean years of schooling). Years since take-off is negative before the internet-user share first reaches 10%.

Linear checks, timing and diagnostics

Algorithm and further simulation results

References

Acemoglu D, Aghion P, Zilibotti F (2006) Distance to frontier, selection, and economic growth. *Journal of the European Economic Association* 4(1):37–74. <https://doi.org/10.1162/jeea.2006.4.1.37>

Table C4 Linear two-way fixed-effects checks and timing of the association.

Specification	Coefficient	(s.e.)
Baseline	-2.84***	(0.79)
Country-specific linear trends	-3.71***	(0.65)
Country-specific quadratic trends	-4.43***	(1.24)
Adding lagged log productivity	-2.73***	(0.74)
Initial-productivity tercile $\times$ year effects	-2.74***	(1.03)
Region $\times$ year effects	-2.26**	(0.88)
Period 1996–2011	-4.35***	(1.01)
Period 2012–2025	0.70	(1.46)
Excluding transition economies	-1.83**	(0.81)
Outcome: GDP growth	-3.83***	(0.78)
Outcome: employment growth	-0.99***	(0.33)
Outcome: growth of employment/population	-0.71**	(0.31)
<i>Timing: TWFE, adoption share dated s (common sample, 2,848 obs.)</i>		
Share in $t - 6$	-1.81**	(0.86)
Share in $t - 5$	-2.24***	(0.84)
Share in $t - 4$	-2.76***	(0.85)
Share in $t - 3$	-2.84***	(0.86)
Share in $t - 2$	-3.23***	(0.87)
Share in $t - 1$	-3.24***	(0.87)
Share in t	-3.18***	(0.91)
Share in $t + 1$	-2.92***	(0.95)
Share in $t + 2$	-2.41**	(0.96)
Share in $t + 3$	-1.81*	(0.99)
Share in $t + 4$	-1.18	(1.04)
FE-DML, lead D_{t+1} as the only treatment	-3.63***	(1.29)
TWFE, lagged change $D_t - D_{t-1}$	-0.37	(1.84)
FE-DML, lagged change $D_t - D_{t-1}$	-2.30	(1.76)

Notes: coefficients per unit of adoption (0–1) on the lagged internet-user share D_t (internet share in $t - 1$), baseline WDI sample and controls, country and year effects, standard errors clustered by country. Trends are country-specific polynomials in the year. Employment growth is GDP growth minus labour-productivity growth; employment/population growth is GDP-per-capita growth minus labour-productivity growth. The timing rows date the share from $t - 6$ to $t + 4$, each in a separate regression on the subsample where all dates are observed. *, **, ***: 10%, 5%, 1%.

Akerman A, Gaarder I, Mogstad M (2015) The skill complementarity of broadband internet. The Quarterly Journal of Economics 130(4):1781–1824. <https://doi.org/10.1093/qje/qjv028>

Angrist N, Djankov S, Goldberg PK, et al (2021) Measuring human capital using global learning data. Nature 592(7854):403–408. <https://doi.org/10.1038/s41586-021-03323-7>

Athey S, Tibshirani J, Wager S (2019) Generalized random forests. The Annals of Statistics 47(2):1148–1178. <https://doi.org/10.1214/18-aos1709>

Table C5 Nuisance fit and sensitivity to the learner’s hyper-parameters (baseline).

<i>Panel A. Out-of-fold R^2, five folds by country</i>								
Learner	Nuisance		R^2	Within-country R^2				
LightGBM	outcome		0.20	0.03				
LightGBM	treatment		0.96	0.77				
Random forest	outcome		0.15	0.03				
Random forest	treatment		0.95	0.74				
Linear (CRE, year dummies)	outcome		0.19	-0.02				
Linear (CRE, year dummies)	treatment		0.88	0.05				

<i>Panel B. LightGBM hyper-parameters (ten splits)</i>								
Moderator	Learning rate	Leaves	Trees	FE-DML	(s.e.)	High – low	(s.e.)	$\hat{\lambda}$
A	0.03	15	400	-2.71	(1.29)	-0.47	(3.53)	178
A	0.01	15	1200	-2.67	(1.33)	-0.68	(3.62)	178
A	0.1	15	150	-2.50	(1.28)	-0.09	(3.36)	10^4
A	0.03	7	400	-2.23	(1.34)	-0.60	(3.63)	0.0178
A	0.03	31	400	-2.94	(1.28)	-0.51	(3.53)	10^4
B	0.03	15	400	-2.71	(1.29)	-0.67	(3.23)	1
B	0.01	15	1200	-2.67	(1.33)	-0.65	(3.30)	1
B	0.1	15	150	-2.50	(1.28)	-0.52	(3.17)	0.562
B	0.03	7	400	-2.23	(1.34)	-0.81	(3.37)	3.16
B	0.03	31	400	-2.94	(1.28)	0.50	(3.03)	0.562

Notes: Panel A: features as in the baseline (controls, their country means, the country means of the treatment and the moderator, the year); the linear benchmark is OLS on the same features with year dummies. Within-country R^2 is computed after two-way demeaning of the outcome and of the out-of-fold residual. Panel B: A = schooling, B = initial productivity; the first row is the baseline learner.

Table C6 Agreement of productivity growth between WDI and PWT on common country-years.

PWT release	Obs.	Countries	Correlation of growth			Rank corr. initial prod.	Within corr. by tercile	Within corr.	
			Pooled	Within	Country means			GDP	Employment
PWT 11.0	2,839	108	0.81	0.75	0.94	0.95	0.68 / 0.80 / 0.73	0.95	0.65
PWT 10.0	2,367	105	0.78	0.72	0.93	0.95	0.65 / 0.76 / 0.71	0.94	0.58

Notes: labour-productivity growth in WDI (GDP per person employed) and PWT (rgdpna/emp) on the country-years in both sources. “Within”: correlation after two-way demeaning, the variation used by the estimators. “By tercile”: within correlation by tercile (low/middle/high) of WDI initial productivity. GDP: WDI GDP growth (per-capita growth plus population growth) vs rgdpna ; employment: implied WDI employment growth vs PWT persons engaged.

Banga K, Harbansh P, Singh S (2026) Digitalisation and structural change: Evidence from cross-country analysis. Journal of Digital Economy 5:314–329. <https://doi.org/10.1016/j.jdec.2026.03.003>

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Benhabib J, Spiegel MM (1994) The role of human capital in economic development: Evidence from aggregate cross-country data. Journal of Monetary Economics 34(2):143–173. [https://doi.org/10.1016/0304-3932\(94\)90047-7](https://doi.org/10.1016/0304-3932(94)90047-7)

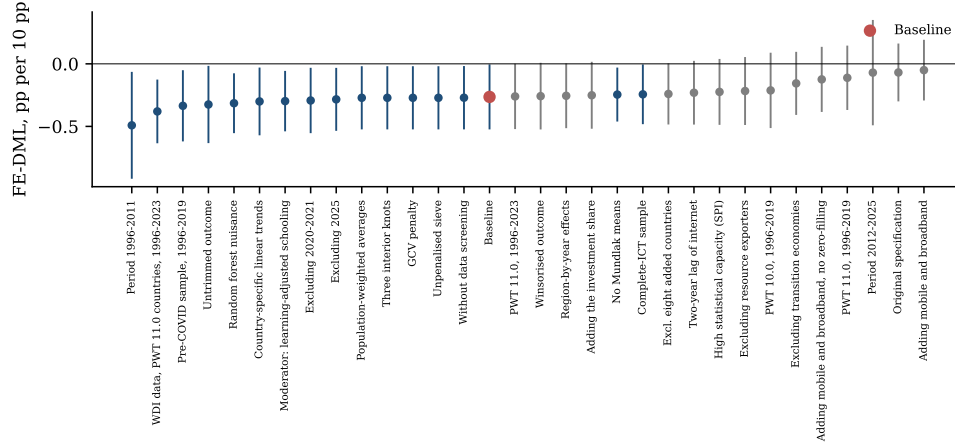


Fig. C1 Specification curve: FE-DML estimate of the average association (pp per ten percentage points of the internet-user share) with 95% country-clustered intervals, for every schooling-moderator specification with the baseline outcome and treatment in Tables 4 and C11, ordered by the estimate. Blue: interval excludes zero; grey: interval includes zero; red: baseline.

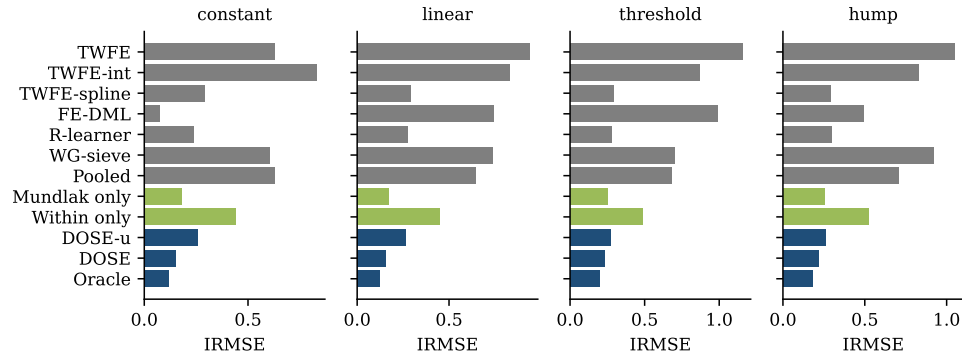


Fig. C2 IRMSE of the estimated effect function by design (stylised design, 100 replications). DOSE denotes Panel-DOSE and DOSE-u its unpenalised version; green bars are ablations; Oracle is the infeasible latent-information benchmark.

Bloom N, Sadun R, Van Reenen J (2012) Americans do IT better: US multinationals and the productivity miracle. *American Economic Review* 102(1):167–201. <https://doi.org/10.1257/aer.102.1.167>

Breiman L (2001) Random forests. *Machine Learning* 45(1):5–32

Bresnahan TF, Brynjolfsson E, Hitt LM (2002) Information technology, workplace organization, and the demand for skilled labor: Firm-level evidence.

Algorithm 1 Panel-DOSE

Require: Panel $(Y_{it}, D_{it}, X_{it}, Z_{it})$; learner $\mathcal{L}$; folds L ; repetitions S ; knots q

- 1: Build $F_{it} = (X_{it}, \bar{X}_i, \bar{D}_i, \bar{Z}_i, t)$ and the cubic B-spline basis $b(\cdot)$ with q interior knots at quantiles of Z
- 2: **for** $s = 1, \dots, S$ **do**
- 3: Randomly partition the *countries* into L folds $I_1, \dots, I_L$
- 4: **for** $l = 1, \dots, L$ **do**
- 5: Fit $\hat{\ell}^{(-l)}$ and $\hat{\mu}^{(-l)}$ by $\mathcal{L}$ on countries not in I_l
- 6: For $i \in I_l$: $\hat{e}_{it}^Y = Y_{it} - \hat{\ell}^{(-l)}(F_{it})$, $\hat{e}_{it}^\Phi = b(Z_{it})\{D_{it} - \hat{\mu}^{(-l)}(F_{it})\}$
- 7: **end for**
- 8: Two-way demean: $\tilde{Y} = \mathcal{M}\hat{e}^Y$, $\tilde{\Phi} = \mathcal{M}\hat{e}^\Phi$
- 9: Compute the leave-countries-out cross-validation criterion $CV_s(\lambda)$ on the grid
- 10: **end for**
- 11: Choose $\hat{\lambda} = \arg \min_{\lambda} \sum_s CV_s(\lambda)$; for each s compute $\hat{\beta}_s$ from (3) and $\hat{V}_s$ from (4) with $\hat{\lambda}$
- 12: For each reported vector of functionals $L\beta$ (including the centred or detrended grids used in the tests): estimate $\hat{c} = \text{med}_s L\hat{\beta}_s$ and covariance $\text{med}_s \{L\hat{V}_s L' + (L\hat{\beta}_s - \hat{c})(L\hat{\beta}_s - \hat{c})'\}$, element by element
- 13: **return** $\hat{\theta}(z)$ with pointwise and simultaneous bands, group averages, BLP slope and shape tests

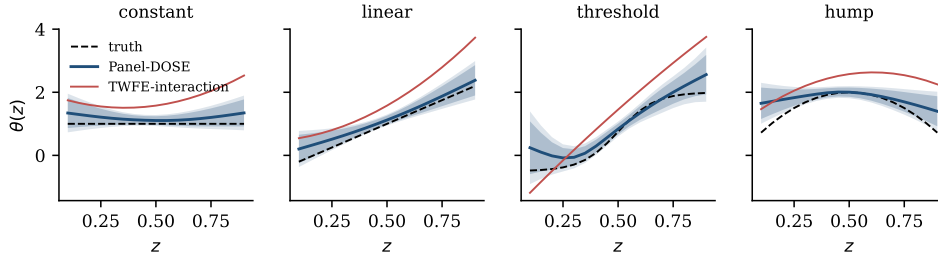


Fig. C3 One simulated sample per design. Dashed black: true $\theta(z)$; blue: Panel-DOSE with 95% pointwise (dark) and simultaneous (light) bands; red: TWFE with quadratic interaction.

The Quarterly Journal of Economics 117(1):339–376. <https://doi.org/10.1162/003355302753399526>

Brynjolfsson E, Hitt LM (2000) Beyond computation: Information technology, organizational transformation and business performance. Journal of Economic Perspectives 14(4):23–48. <https://doi.org/10.1257/jep.14.4.23>

Brynjolfsson E, Rock D, Syverson C (2021) The productivity J-curve: How intangibles complement general purpose technologies. American Economic Journal: Macroeconomics 13(1):333–372. <https://doi.org/10.1257/mac.20180386>

Table C7 Stylised Monte Carlo design: coverage of nominal 95% intervals.

Estimator	Pointwise	Simultaneous	Average	High-low	BLP slope
<i>Design: constant $\theta(z)$</i>					
TWFE-interaction	0.04 (0.01)	0.00 (0.00)	0.00 (0.00)	0.17 (0.04)	0.17 (0.04)
TWFE-spline	0.63 (0.02)	0.31 (0.05)	0.21 (0.04)	0.93 (0.03)	0.90 (0.03)
FE-DML (constant)	0.91 (0.03)	0.90 (0.03)	0.91 (0.03)	1.00 (0.00)	0.91 (0.03)
WG-DML sieve	0.24 (0.02)	0.00 (0.00)	0.00 (0.00)	0.80 (0.04)	0.71 (0.05)
Pooled DML sieve	0.11 (0.01)	0.00 (0.00)	0.00 (0.00)	0.91 (0.03)	0.92 (0.03)
Ablation: Mundlak only	0.86 (0.02)	0.78 (0.04)	0.79 (0.04)	0.96 (0.02)	0.95 (0.02)
Ablation: within only	0.26 (0.02)	0.04 (0.02)	0.01 (0.01)	0.96 (0.02)	0.96 (0.02)
Panel-DOSE (unpenalised)	0.93 (0.01)	0.88 (0.03)	0.90 (0.03)	0.96 (0.02)	0.93 (0.03)
Panel-DOSE	0.91 (0.02)	0.84 (0.04)	0.92 (0.03)	0.97 (0.02)	0.94 (0.02)
Latent-information oracle	0.92 (0.01)	0.85 (0.04)	0.95 (0.02)	0.93 (0.03)	0.93 (0.03)
<i>Design: linear $\theta(z)$</i>					
TWFE-interaction	0.04 (0.01)	0.00 (0.00)	0.00 (0.00)	0.17 (0.04)	0.17 (0.04)
TWFE-spline	0.63 (0.02)	0.31 (0.05)	0.21 (0.04)	0.93 (0.03)	0.90 (0.03)
FE-DML (constant)	0.14 (0.00)	0.00 (0.00)	0.98 (0.01)	0.00 (0.00)	0.00 (0.00)
WG-DML sieve	0.37 (0.02)	0.02 (0.01)	0.10 (0.03)	0.34 (0.05)	0.23 (0.04)
Pooled DML sieve	0.12 (0.01)	0.00 (0.00)	0.00 (0.00)	0.92 (0.03)	0.92 (0.03)
Ablation: Mundlak only	0.90 (0.02)	0.87 (0.03)	0.87 (0.03)	0.94 (0.02)	0.92 (0.03)
Ablation: within only	0.25 (0.02)	0.05 (0.02)	0.02 (0.01)	0.89 (0.03)	0.84 (0.04)
Panel-DOSE (unpenalised)	0.94 (0.01)	0.86 (0.03)	0.93 (0.03)	0.95 (0.02)	0.91 (0.03)
Panel-DOSE	0.89 (0.02)	0.85 (0.04)	0.89 (0.03)	0.93 (0.03)	0.90 (0.03)
Latent-information oracle	0.92 (0.01)	0.85 (0.04)	0.95 (0.02)	0.93 (0.03)	0.93 (0.03)
<i>Design: threshold $\theta(z)$</i>					
TWFE-interaction	0.16 (0.00)	0.00 (0.00)	0.00 (0.00)	0.03 (0.02)	0.00 (0.00)
TWFE-spline	0.63 (0.02)	0.31 (0.05)	0.21 (0.04)	0.93 (0.03)	0.90 (0.03)
FE-DML (constant)	0.07 (0.00)	0.00 (0.00)	0.99 (0.01)	0.00 (0.00)	0.00 (0.00)
WG-DML sieve	0.43 (0.02)	0.04 (0.02)	0.05 (0.02)	0.49 (0.05)	0.44 (0.05)
Pooled DML sieve	0.18 (0.01)	0.00 (0.00)	0.00 (0.00)	0.92 (0.03)	0.91 (0.03)
Ablation: Mundlak only	0.80 (0.02)	0.49 (0.05)	0.89 (0.03)	0.94 (0.02)	0.95 (0.02)
Ablation: within only	0.36 (0.02)	0.03 (0.02)	0.01 (0.01)	0.91 (0.03)	0.87 (0.03)
Panel-DOSE (unpenalised)	0.93 (0.01)	0.83 (0.04)	0.88 (0.03)	0.96 (0.02)	0.90 (0.03)
Panel-DOSE	0.81 (0.02)	0.55 (0.05)	0.88 (0.03)	0.90 (0.03)	0.88 (0.03)
Latent-information oracle	0.86 (0.02)	0.71 (0.05)	0.96 (0.02)	0.95 (0.02)	0.87 (0.03)
<i>Design: hump $\theta(z)$</i>					
TWFE-interaction	0.04 (0.01)	0.00 (0.00)	0.00 (0.00)	0.17 (0.04)	0.17 (0.04)
TWFE-spline	0.63 (0.02)	0.31 (0.05)	0.21 (0.04)	0.93 (0.03)	0.90 (0.03)
FE-DML (constant)	0.30 (0.01)	0.00 (0.00)	0.73 (0.04)	0.00 (0.00)	0.00 (0.00)
WG-DML sieve	0.08 (0.01)	0.00 (0.00)	0.00 (0.00)	0.94 (0.02)	0.94 (0.02)
Pooled DML sieve	0.11 (0.02)	0.00 (0.00)	0.00 (0.00)	0.93 (0.03)	0.92 (0.03)
Ablation: Mundlak only	0.85 (0.02)	0.68 (0.05)	0.69 (0.05)	0.95 (0.02)	0.95 (0.02)
Ablation: within only	0.27 (0.02)	0.03 (0.02)	0.00 (0.00)	0.92 (0.03)	0.93 (0.03)
Panel-DOSE (unpenalised)	0.94 (0.01)	0.87 (0.03)	0.93 (0.03)	0.96 (0.02)	0.92 (0.03)
Panel-DOSE	0.88 (0.01)	0.69 (0.05)	0.91 (0.03)	0.95 (0.02)	0.94 (0.02)
Latent-information oracle	0.90 (0.01)	0.73 (0.04)	0.97 (0.02)	0.93 (0.03)	0.93 (0.03)

Notes: 100 replications; Monte Carlo standard errors in parentheses. “Pointwise” is the share of grid points at which the interval covers $\theta(z)$; “Simultaneous” the share of replications in which the sup- t band covers the whole function; “Average”, “High-low” and “BLP slope” refer to the sample-average effect, the tercile contrast and the best linear projection slope. The constant-effect FE-DML estimator targets the variance-weighted average, so its coverage of the other quantities is reported only to show the cost of assuming a constant effect.

Byrne DM, Fernald JG, Reinsdorf MB (2016) Does the United States have a productivity slowdown or a measurement problem? Brookings Papers on Economic Activity 2016(1):109–182. <https://doi.org/10.1353/eca.2016.0014>

Table C8 Stylised design, threshold θ : sensitivity to sample size and confounding.

	$G = 50$ $\rho = 1$	$G = 100$ $\rho = 0.25$	$G = 100$ $\rho = 0.5$	$G = 100$ $\rho = 1$	$G = 100$ $\rho = 2$	$G = 200$ $\rho = 1$
<i>IRMSE</i>						
TWFE-interaction	0.89	0.46	0.56	0.87	1.66	0.87
TWFE-spline	0.37	0.17	0.20	0.29	0.52	0.25
FE-DML + R-learner (tuned)	0.36	0.27	0.28	0.28	0.33	0.21
Panel-DOSE (unpenalised)	0.43	0.26	0.27	0.27	0.34	0.18
Panel-DOSE	0.35	0.22	0.23	0.23	0.29	0.15
<i>Pointwise coverage</i>						
TWFE-interaction	0.20 (0.01)	0.25 (0.02)	0.12 (0.01)	0.16 (0.00)	0.16 (0.01)	0.11 (0.00)
Panel-DOSE (unpenalised)	0.92 (0.01)	0.93 (0.01)	0.92 (0.01)	0.93 (0.01)	0.91 (0.02)	0.91 (0.01)
Panel-DOSE	0.82 (0.02)	0.81 (0.03)	0.80 (0.03)	0.81 (0.02)	0.73 (0.02)	0.89 (0.01)
<i>Simultaneous coverage</i>						
TWFE-interaction	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Panel-DOSE (unpenalised)	0.82 (0.05)	0.76 (0.06)	0.76 (0.06)	0.83 (0.04)	0.84 (0.05)	0.86 (0.05)
Panel-DOSE	0.52 (0.07)	0.54 (0.07)	0.54 (0.07)	0.55 (0.05)	0.40 (0.07)	0.70 (0.07)
<i>Coverage of high-low contrast</i>						
TWFE-interaction	0.22 (0.06)	0.22 (0.06)	0.10 (0.04)	0.03 (0.02)	0.04 (0.03)	0.00 (0.00)
Panel-DOSE (unpenalised)	0.94 (0.03)	0.92 (0.04)	0.96 (0.03)	0.96 (0.02)	1.00 (0.00)	0.94 (0.03)
Panel-DOSE	0.96 (0.03)	0.92 (0.04)	0.88 (0.05)	0.90 (0.03)	0.94 (0.03)	0.94 (0.03)

Notes: 50 replications per cell except $G = 100, \rho = 1$ (100). Monte Carlo standard errors of coverage rates in parentheses. ρ scales the nonlinear confounding in the outcome equation.

Table C9 Plasmode design: linear TWFE benchmarks.

True θ , estimator	Bias	RMSE	Coverage	Rejection
<i>Moderator: Schooling</i>				
no effect, TWFE-interaction	-0.67	1.25	0.90 (0.02)	0.10 (0.02)
no effect, TWFE-spline	1.06	2.27	0.91 (0.02)	0.10 (0.02)
linear gradient, TWFE-interaction	-0.60	1.15	0.94 (0.02)	1.00 (0.00)
linear gradient, TWFE-spline	1.10	2.25	0.90 (0.03)	1.00 (0.00)
threshold, TWFE-interaction	-0.84	1.29	0.94 (0.02)	1.00 (0.00)
threshold, TWFE-spline	1.18	2.29	0.90 (0.03)	1.00 (0.00)
<i>Moderator: Initial productivity</i>				
no effect, TWFE-interaction	0.59	1.10	0.89 (0.02)	0.11 (0.02)
no effect, TWFE-spline	1.05	2.13	0.68 (0.03)	0.33 (0.03)
linear gradient, TWFE-interaction	0.60	1.15	0.85 (0.04)	1.00 (0.00)
linear gradient, TWFE-spline	1.13	2.28	0.62 (0.05)	1.00 (0.00)
threshold, TWFE-interaction	0.01	0.98	0.95 (0.02)	1.00 (0.00)
threshold, TWFE-spline	1.11	2.26	0.62 (0.05)	1.00 (0.00)

Notes: see Table 2. TWFE with a linear interaction or with additive spline controls and a spline in the moderator; tercile contrast only. Years since take-off is omitted from the linear benchmarks.

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Table C10 Penalty, identifying variation and treatment support.

Specification	$\hat{\lambda}$	edf	Residual share	Residual share by tercile	Median D by tercile	Within-country s.d. of D	Countries by tercile
A: Baseline	562	2.0	0.22	0.11 / 0.30 / 0.30	0.04 / 0.30 / 0.74	0.10 / 0.16 / 0.19	76 / 96 / 72
A: Common sample	31.6	2.5	0.23	0.10 / 0.34 / 0.31	0.05 / 0.30 / 0.75	0.10 / 0.16 / 0.18	74 / 92 / 70
B: Baseline	1	4.0	0.22	0.14 / 0.34 / 0.23	0.07 / 0.31 / 0.68	0.17 / 0.26 / 0.31	53 / 49 / 47
B: Common sample	10^4	2.0	0.23	0.14 / 0.36 / 0.24	0.07 / 0.33 / 0.69	0.17 / 0.27 / 0.31	51 / 46 / 44
C: Baseline	0.0316	5.9	0.18	0.05 / 0.37 / 0.37	0.02 / 0.30 / 0.80	0.03 / 0.13 / 0.08	142 / 149 / 126

Notes: $\hat{\lambda}$: cross-validated penalty (median over splits; 10^4 is the upper end of the grid, an effectively linear fit); edf: effective degrees of freedom. “Residual share”: variance of the two-way demeaned residual $D - \hat{\mu}$ relative to the variance of two-way demeaned D , overall and by tercile (low/middle/high) of the moderator. Countries by tercile can overlap for time-varying moderators.

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Table C11 Further robustness checks: schooling moderator.

Specification	Average	Tercile of moderator			High –	BLP	Constancy
	(FE-DML)	Low	Middle	High	low	slope	
<i>Panel A. Moderator: mean years of schooling</i>							
Excluding the eight countries outside the complete-ICT sample [†]	–2.41* (1.25)	1.22 (2.65)	–2.93** (1.39)	–3.00* (1.64)	–4.03 (3.34)	–0.60 (0.46)	0.28
Region-by-year effects [†]	–2.55* (1.32)	–2.15 (3.07)	–2.52* (1.43)	–2.54 (1.67)	–0.33 (3.68)	–0.05 (0.50)	0.97
Original specification [†]	–0.69 (1.18)	–2.34 (2.24)	–0.94 (1.20)	0.36 (1.42)	2.67 (2.88)	0.36 (0.39)	0.18
Adding the investment share [†]	–2.51* (1.36)	–1.70 (3.03)	–2.35* (1.40)	–3.24* (1.75)	–1.54 (3.71)	–0.21 (0.50)	0.70
Adding mobile and broadband, no zero-filling [†]	–1.24 (1.33)	–4.36* (2.29)	–1.19 (1.41)	2.21 (1.51)	6.44** (3.00)	0.85** (0.43)	0.08
Treatment: composite ICT index [†]	–3.03* (1.56)	–1.38 (3.78)	–2.60 (1.78)	–3.94** (1.81)	–2.13 (4.52)	–0.29 (0.61)	0.68
Random forest nuisance [†]	–3.15*** (1.22)	–3.04 (2.31)	–3.15** (1.25)	–2.99* (1.53)	0.19 (2.89)	0.03 (0.39)	0.96
Unpenalised sieve [†]	–2.71** (1.29)	–1.39 (2.93)	–2.57 (1.89)	–1.46 (1.45)	–0.39 (3.22)	–0.31 (0.46)	0.53
Three interior knots [†]	–2.71** (1.29)	–2.26 (2.78)	–2.77** (1.40)	–2.61 (1.68)	–0.50 (3.53)	–0.08 (0.48)	0.91
GCV penalty [†]	–2.71** (1.29)	–1.44 (2.98)	–2.54 (1.88)	–1.85 (1.50)	–0.49 (3.27)	–0.31 (0.47)	0.56
Two-year lag of internet [†]	–2.31* (1.30)	–2.33 (3.21)	–2.26 (1.48)	–2.12 (1.53)	–0.16 (3.89)	–0.02 (0.53)	0.99
Pre-COVID sample, 1996–2019 [†]	–3.35** (1.45)	–1.90 (3.59)	–2.96* (1.68)	–3.76** (1.90)	–1.78 (4.39)	–0.24 (0.60)	0.69
Excluding 2020–2021 [†]	–2.93** (1.33)	–3.03 (3.10)	–2.90** (1.43)	–2.43 (1.68)	0.43 (3.89)	0.06 (0.53)	0.91
Excluding 2025 [†]	–2.84** (1.28)	–2.32 (2.68)	–2.84** (1.34)	–3.07* (1.69)	–1.14 (3.55)	–0.15 (0.48)	0.76
Untrimmed outcome [†]	–3.25** (1.58)	–2.97 (3.17)	–3.26** (1.64)	–3.46 (2.17)	–0.12 (4.21)	–0.02 (0.57)	1.00
Winsorised outcome [†]	–2.58* (1.36)	–2.18 (2.96)	–2.53* (1.41)	–2.75 (1.75)	–0.68 (3.79)	–0.09 (0.52)	0.88
Without data screening [†]	–2.71** (1.29)	–1.91 (2.86)	–2.66* (1.36)	–2.80* (1.62)	–1.05 (3.56)	–0.14 (0.48)	0.79
Excluding resource exporters [†]	–2.17 (1.39)	–2.72 (2.42)	–2.20 (1.38)	–1.89 (1.92)	0.68 (3.38)	0.09 (0.46)	0.70
High statistical capacity (SPI) [†]	–2.24* (1.34)	–0.58 (2.63)	–3.66** (1.57)	–1.15 (1.94)	–0.28 (3.24)	–0.00 (0.65)	0.13
Outcome: GDP per capita growth [†]	–3.02** (1.36)	–3.09 (2.71)	–3.08** (1.39)	–2.92 (1.89)	0.44 (3.68)	0.06 (0.50)	0.93
Population-weighted averages [†]	–2.71** (1.29)	–2.33 (2.78)	–2.76** (1.38)	–2.65 (1.70)	–0.47 (3.53)	–0.07 (0.48)	0.92
No Mundlak means [†]	–2.46** (1.10)	–1.84 (2.02)	–2.57** (1.19)	–2.66* (1.40)	–0.91 (2.55)	–0.14 (0.34)	0.59
WDI data, PWT 11.0 countries, 1996–2023	–3.80*** (1.30)	–7.86** (3.09)	–4.07*** (1.35)	–1.74 (1.72)	6.29 (3.92)	0.90 (0.56)	0.13
Moderator: learning-adjusted schooling [†]	–2.98** (1.23)	–3.08 (2.88)	–2.96** (1.38)	–2.89** (1.36)	0.44 (3.46)	0.08 (0.59)	0.96
PWT 11.0, 1996–2019 [†]	–1.11 (1.31)	–4.71 (5.49)	–0.16 (1.72)	–1.21 (1.22)	3.33 (5.87)	2.90 (4.25)	0.45
PWT 11.0: outcome TFP growth [†]	–2.71** (1.37)	–9.40** (4.35)	–2.45 (1.76)	–1.15 (1.36)	8.46* (4.63)	7.32** (3.66)	0.26

Notes: see Table 4. †: ten repetitions of the sample split (20 otherwise). “GCV penalty” selects λ by generalised cross-validation. “Population-weighted averages” changes only the weighting of the sample-average effect, which is reported in the text. “No zero-filling” leaves broadband missing before a country’s first reported value instead of setting it to zero.

Table C12 Further robustness checks: initial productivity.

Specification	Average	Tercile of moderator			High –	BLP	Constancy
	(FE-DML)	Low	Middle	High	low	slope	test p
Panel B. Moderator: initial log GDP per worker							
Excluding the eight countries outside the complete-ICT sample [†]	–2.41* (1.25)	–2.02 (2.60)	–2.60* (1.35)	–2.56** (1.30)	–0.38 (2.93)	–0.17 (1.16)	0.74
Region-by-year effects [†]	–2.55* (1.32)	–0.97 (2.91)	–4.75*** (1.77)	–0.98 (1.51)	0.19 (3.25)	–0.29 (1.36)	0.21
Original specification [†]	–0.69 (1.21)	0.23 (2.23)	–1.68 (1.81)	0.27 (1.53)	–0.21 (2.80)	–1.21 (1.15)	0.12
Adding the investment share [†]	–2.51* (1.36)	–2.40 (2.83)	–3.27* (1.69)	–2.03 (1.46)	0.79 (3.33)	0.13 (1.34)	0.43
Adding mobile and broadband [†]	–0.50 (1.24)	0.48 (2.16)	–1.97 (2.05)	0.64 (1.43)	0.01 (2.60)	–1.22 (1.03)	0.08
Adding mobile and broadband, no zero-filling [†]	–1.24 (1.33)	–1.35 (2.08)	–2.78 (2.12)	–0.80 (2.02)	0.83 (2.94)	0.18 (1.17)	0.05
Treatment: composite ICT index [†]	–3.03* (1.56)	–6.45** (3.01)	–3.72** (1.71)	–0.52 (1.43)	5.54* (3.31)	2.12 (1.33)	0.09
Random forest nuisance [†]	–3.15*** (1.22)	–2.06 (2.35)	–4.76*** (1.54)	–2.05 (1.41)	0.28 (2.75)	–0.12 (1.08)	0.16
Unpenalised sieve [†]	–2.71** (1.29)	–3.41 (3.30)	–6.45*** (1.95)	0.80 (1.42)	3.74 (3.94)	1.89 (1.80)	0.05
Three interior knots [†]	–2.71** (1.29)	–1.03 (2.88)	–4.91*** (1.76)	–1.47 (1.39)	–0.70 (3.22)	–0.72 (1.31)	0.11
GCV penalty [†]	–2.71** (1.29)	–2.14 (3.44)	–6.39*** (1.94)	0.60 (1.42)	2.35 (3.97)	1.18 (1.82)	0.06
Two-year lag of internet [†]	–2.31* (1.30)	–0.76 (2.75)	–2.71* (1.48)	–2.57* (1.48)	–2.06 (3.26)	–0.89 (1.30)	0.40
Pre-COVID sample, 1996–2019 [†]	–3.35** (1.45)	–4.21 (4.18)	–3.60* (1.89)	–3.01** (1.34)	0.88 (4.58)	0.35 (1.84)	0.86
Excluding 2020–2021 [†]	–2.93** (1.33)	–2.47 (2.61)	–3.48** (1.53)	–2.35 (1.47)	0.13 (3.16)	–0.06 (1.26)	0.32
Excluding 2025 [†]	–2.84** (1.28)	–2.36 (2.69)	–3.86** (1.53)	–2.25* (1.32)	–0.41 (3.17)	–0.31 (1.28)	0.29
Untrimmed outcome [†]	–3.25** (1.58)	–3.55 (3.42)	–4.60** (1.96)	–2.05* (1.21)	1.57 (3.66)	0.43 (1.46)	0.19
Winsorised outcome [†]	–2.58* (1.36)	–1.69 (2.91)	–4.43* (1.84)	–1.27 (1.36)	0.76 (3.33)	0.01 (1.35)	0.23
Without data screening [†]	–2.71** (1.29)	–0.68 (2.77)	–4.84*** (1.71)	–1.47 (1.36)	–0.58 (3.21)	–0.61 (1.28)	0.09
Excluding resource exporters [†]	–2.17 (1.39)	–0.97 (2.41)	–4.00* (2.06)	–0.81 (1.37)	0.04 (2.88)	–0.23 (1.20)	0.26
High statistical capacity (SPI) [†]	–2.24* (1.34)	–2.47 (2.53)	–2.39 (1.76)	–0.37 (1.26)	2.09 (2.84)	0.32 (1.78)	0.08
Outcome: GDP per capita growth [†]	–3.02** (1.36)	–1.18 (2.70)	–4.15** (1.69)	–2.78* (1.49)	–1.79 (3.05)	–0.94 (1.25)	0.34
Population-weighted averages [†]	–2.71** (1.29)	–1.01 (2.89)	–4.97*** (1.76)	–1.44 (1.40)	–0.67 (3.23)	–0.72 (1.32)	0.13
No Mundlak means [†]	–2.46** (1.10)	–0.50 (2.21)	–5.38*** (1.70)	0.27 (1.30)	0.46 (2.54)	–1.16 (1.23)	<0.01
Moderator: distance to US frontier [†]	–1.98 (1.28)	3.82 (2.66)	–4.12** (1.98)	–1.09 (1.22)	–5.16* (2.88)	–3.17** (1.36)	0.11
Countries with pre-sample productivity only [†]	–2.72** (1.35)	–1.42 (2.74)	–4.81*** (1.79)	–1.29 (1.42)	0.00 (3.20)	–0.31 (1.35)	0.24
PWT 11.0, 1996–2019 [†]	–1.11 (1.31)	–6.25 (4.16)	–1.96 (1.62)	1.08 (1.55)	7.73 (4.88)	3.45 (2.18)	0.12
PWT 11.0: outcome TFP growth [†]	–2.71** (1.37)	–5.48 (3.50)	–5.91*** (2.05)	1.09 (1.56)	6.56* (3.71)	3.00 (2.33)	0.02
PWT 11.0, national-prices moderator [†]	–2.22* (1.26)	–7.77*** (3.02)	–2.00 (1.41)	0.47 (1.32)	8.21** (3.42)	3.60** (1.51)	0.08
PWT 10.0, national-prices moderator [†]	–2.02 (1.58)	–10.62*** (3.33)	–3.39* (1.45)	1.84 (1.72)	12.21*** (4.17)	5.04*** (1.72)	0.01

Notes: see Table 4. [†]: ten repetitions of the sample split (20 otherwise). “GCV penalty” selects λ by generalised cross-validation. “Population-weighted averages” changes only the weighting of the sample-average effect, which is reported in the text. “No zero-filling” leaves broadband missing before a country’s first reported value instead of setting it to zero.

Table C13 Further robustness checks: years since take-off.

Specification	Average	Tercile of moderator			High –	BLP	Constancy
	(FE-DML)	Low	Middle	High	low	slope	
<i>Panel C. Moderator: years since internet take-off (share ≥ 10%)</i>							
Country-specific linear trends [†]	–0.84 (1.37)	–4.24 (3.77)	–1.59 (1.61)	0.95 (2.15)	5.13 (5.26)	0.22 (0.22)	0.34
Excluding transition economies [†]	–1.35 (1.35)	–1.54 (3.28)	–1.37 (1.64)	–1.69 (1.76)	–0.08 (4.12)	–0.00 (0.17)	0.98
Excluding the eight countries outside the complete-ICT sample [†]	–1.64 (1.32)	–9.11* (4.93)	–3.02* (1.59)	0.24 (1.65)	9.51* (5.44)	0.39* (0.23)	0.15
Period 1996–2011 [†]	–3.04 (2.28)	10.94 (23.07)	–5.94 (5.27)	–2.75 (2.32)	–13.27 (23.27)	–0.90 (1.52)	0.73
Period 2012–2025 [†]	–0.62 (1.98)	–0.96 (3.35)	–0.65 (1.93)	–0.61 (3.15)	0.64 (5.18)	0.04 (0.30)	0.90
Region-by-year effects [†]	–1.62 (1.38)	1.97 (10.38)	–2.74* (1.64)	–0.36 (1.70)	–2.71 (10.55)	–0.28 (0.53)	0.17
Random forest nuisance [†]	–1.64 (1.40)	7.48 (13.27)	–2.99* (1.66)	–0.80 (1.75)	–8.00 (13.31)	–0.77 (0.72)	0.06
Pre-COVID sample, 1996–2019 [†]	–1.70 (1.72)	–7.08 (4.39)	–3.65 (2.26)	–0.30 (1.98)	7.04 (5.09)	0.35 (0.25)	0.17
Adding mobile and broadband [†]	–1.34 (1.46)	–2.71 (3.70)	–1.52 (1.86)	–0.77 (1.73)	2.45 (4.22)	0.10 (0.18)	0.57

Notes: see Table 4. [†]: ten repetitions of the sample split (20 otherwise). “GCV penalty” selects λ by generalised cross-validation. “Population-weighted averages” changes only the weighting of the sample-average effect, which is reported in the text. “No zero-filling” leaves broadband missing before a country’s first reported value instead of setting it to zero.